Entanglement Reachability and Structural Confinement Under Local Quantum Dynamics: An Axiomatic Approach With Physical Realizations

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Abstract

We study entanglement reachability in quantum many-body systems evolving under physically admissible local dynamics. While Lieb–Robinson bounds constrain the propagation of information, and complexity-based approaches characterize the resources required to represent or prepare quantum states, these frameworks do not by themselves determine which entanglement structures are dynamically attainable. The central question addressed in this work is therefore: which entanglement structures can arise under constrained quantum evolution, and which remain inaccessible despite being kinematically allowed?

We introduce a unified axiomatic framework for entanglement reachability based on the notion of *structural confinement*. The framework incorporates finite-range locality, finite-dimensional memory degrees of freedom, and abstract representability constraints expressed through operator subsystems. Rather than characterizing accessibility directly in Hilbert space, we formulate the problem on an induced structural space of entanglement configurations and quantum information patterns.

We establish a structural confinement theorem showing that admissible dynamics generate only a restricted reachable region in structural space. The theorem provides a general criterion for identifying forbidden structures whenever their defining information cannot be preserved within the admissible representational framework. Thus, locality, memory, and representability constraints do not merely limit the rate of information propagation; they impose additional dynamical restrictions on the set of quantum structures that can actually be realized.

The generality of the framework is demonstrated through three explicit realizations involving multipartite coherence, tensor-network representability, and protected logical information in quantum error- correcting codes. Although these realizations involve distinct physical structures, they are governed by the same underlying confinement mechanism: dynamical accessibility is determined by the compatibility between physical evolution and the structural information retained by the chosen representation.

These results establish structural confinement as a general organizing principle for entanglement reachability under constrained quantum dynamics. The framework provides a unified language for deriving dynamical accessibility limits beyond propagation bounds alone and for classifying which quantum structures are reachable, forbidden, or conditionally accessible in locally constrained many-body systems.

1 Introduction

The dynamical accessibility of quantum structures in many-body systems is fundamentally constrained not only by locality and interaction structure, but also by the compatibility between physical evolution and the representational framework used to describe quantum information. Over the past decades, related questions have been studied from multiple complementary perspectives, including Lieb–Robinson bounds on information propagation, circuit complexity and state-preparation depth, tensor-network representability, and the dynamics of scrambling in chaotic quantum systems. These approaches provide partial characterizations of how correlations and entanglement spread, and how efficiently certain quantum states can be prepared under restricted resources.

Recent developments have further refined this landscape in several directions. Lieb–Robinson-type bounds have been extended to long-range interacting systems and open quantum dynamics, while tensor-network methods have been used to characterize efficiently representable subspaces of many-body Hilbert space, including matrix product state (MPS) and projected entangled pair state (PEPS) manifolds. In parallel, quantum information scrambling and out-of-time-order correlators have provided dynamical diagnostics of operator growth, and quantum control theory has addressed state-preparation and controllability questions in local and dissipative settings. More recently, frameworks based on quantum channels, process tensors, and non-Markovian quantum evolution have highlighted the role of memory effects in structured open-system dynamics. Despite these advances, these approaches typically address locality, complexity, and memory effects in isolation or in partially overlapping regimes, rather than as a unified framework for determining which quantum structures remain dynamically accessible under simultaneous physical and representational constraints.

Despite this progress, a unifying structural question remains less developed:

> *Which entanglement structures, or more generally, which quantum structures, are dynamically reachable under physically admissible evolution, and which are fundamentally inaccessible even when they are kinematically consistent with the underlying physical constraints?*

In this question, "structure" is not restricted to a particular entanglement measure or complexity parameter. Rather, it refers to the information retained by a chosen structural description, which may include multipartite coherence, tensor-network representability, logical quantum information, or other operationally meaningful features.

Most existing frameworks focus either on kinematic constraints, such as entanglement entropy bounds and Lieb–Robinson light-cone structure, or on resource-theoretic measures of complexity, such as circuit depth or tensor-network bond dimension. These approaches characterize properties of states, operators, or preparation procedures, but they do not generally provide a unified framework for determining which structural descriptions of quantum states remain dynamically accessible under simultaneous physical and representational constraints.

In this work, we introduce an axiomatic framework for addressing this problem by defining and analyzing the *reachable set of many-body quantum states under admissible dynamics*. The admissible class of dynamics is formulated axiomatically, incorporating (i) finite-range locality of interactions, (ii) finite-dimensional auxiliary memory degrees of freedom with mixing dynamics, and (iii) an abstract notion of representability constraints encoded via operator subsystems.

Within this framework, we define the reachable set of states at time T, denoted $\mathcal{R}(T)$, consisting of all states obtainable from admissible initial conditions under admissible completely positive trace-preserving evolutions, and the induced reachable region in structural space. The latter is the central object of the theory. The confinement theorem is therefore formulated on structural space rather than directly on Hilbert space, reflecting the fact that dynamical accessibility is fundamentally a property of structural information rather than of individual state vectors or density operators. This object provides a dynamical refinement of kinematic state-space constraints, incorporating both finite-speed information propagation and structured non-Markovian feedback mediated by memory degrees of freedom.

More precisely, the primary object of interest is not only the reachable set of states $\mathcal{R}(T)$, but the induced reachable region in a general structural space, obtained after quotienting states according to the structural information relevant to the chosen realization, under the combined constraints of locality, finite memory, and representability.

We further study how entanglement across bipartitions and correlation structures induced on the interaction graph evolve under admissible dynamics, leading to a classification of reachable and forbidden entanglement configurations.

The main conceptual contribution of this work is to introduce *structural confinement* as a general principle governing dynamical accessibility. The central shift is from propagation constraints to representation-dependent attainability constraints. While Lieb–Robinson-type bounds characterize the causal structure of information spreading, they do not determine which structural configurations remain accessible once additional requirements are imposed on the information that must be preserved. We show that locality, memory, and representability constraints combine to produce forbidden regions in structural space that cannot be identified from propagation geometry alone.

Unlike existing no-go results, which are typically tied to a particular physical mechanism or complexity measure, the present framework derives impossibility regions from an abstract compatibility principle between dynamics and structural representation. The resulting confinement mechanism is therefore representation-independent, while its physical interpretation is determined by the chosen structural realization.

We develop this idea along three directions:

First, we establish a structural confinement theorem showing that admissible dynamics generate only a restricted region of structural space. The theorem provides a general criterion for identifying inaccessible structures whenever their defining information exceeds the representability scale preserved by the dynamics.

Second, we demonstrate the generality of the framework through explicit realizations involving multipartite coherence, tensor-network representation, and protected logical information. These examples show that structurally different obstructions arise from the same underlying confinement mechanism.

Third, we establish a conceptual separation between *kinematic existence and dynamical accessibility* on many-body quantum states. A quantum structure may exist within the full kinematic state space while remaining unreachable because its defining information cannot be retained within the admissible representational framework.

Together, these results establish structural confinement as a general organizing principle for quantum dynamical accessibility. The framework shows that the relevant boundary of quantum evolution is not determined solely by what states are allowed by Hilbert-space kinematics, but by which structural features can remain compatible with the physical and representational constraints imposed on the dynamics. Structural confinement therefore provides a general mathematical principle for deriving dynamical accessibility limits across distinct quantum information settings, independent of any particular realization such as multipartite coherence, tensor-network representations, or quantum error-correcting codes.

The remainder of the paper is organized as follows. Section 2 introduces the axiomatic framework defining admissible dynamics. Section 3 formalizes the notion of reachable sets and entanglement structures. Section 4 develops the structural theory of entanglement reachability by introducing induced structural functionals, establishing the structural confinement theorem, and deriving consequences for reachable regions, forbidden entanglement structures, and the separation between kinematic and dynamical constraints. Section 5 provides three explicit realizations of the confinement mechanism. Sections 6 and 7 discuss the conceptual implications, scope, and future extensions of the framework.

1.1 Structural confinement versus existing accessibility bounds

Several established frameworks constrain different aspects of quantum accessibility. Lieb–Robinson bounds restrict the propagation of information under local dynamics [15, 3, 17]. Complexity-based approaches constrain the resources required for state preparation or computation [1, 24]. Tensor-network methods characterize efficient representations of quantum states and operators through restricted ansatz classes [22, 23, 20, 21, 16]. Quantum error-correction theory constrains the preservation and recovery of encoded information under noise [11, 19, 18].

The structural confinement framework addresses a distinct question: which structural sectors remain dynamically reachable when only a finite representable operator subsystem is retained? The obstruction is therefore not a bound on propagation, preparation complexity, representation efficiency, or error correction individually, but a compatibility constraint between dynamics and the operator information required to define a structure.

Framework	Constraint source	Object constrained
Lieb–Robinson theory	locality	information propagation
Complexity theory	computational resources	state preparation
Tensor networks	restricted ansatz	state/operator representation
Quantum error correction	noise and recovery	logical information preservation
Structural confinement	representability compatibility	structural reachability

Thus, the present framework introduces representability as an additional layer of accessibility constraints. A structure may be compatible with locality, exist within the Hilbert space, and admit efficient descriptions in other settings, while remaining dynamically inaccessible because the information defining that structure lies outside the admissible representational sector. The GHZ, tensor-network, and logical-code realizations developed below provide distinct examples of this general mechanism.

2 Axiomatic Dynamical Framework

We formulate the class of admissible quantum many-body dynamics as an axiomatic system incorporating locality, memory, and representability constraints. All results in this work are derived within this framework.

We formulate the dynamics in the Schrödinger picture, where states evolve under completely positive trace-preserving maps. Operator-algebraic structures in the Heisenberg picture are used solely to formulate representability constraints.

Axiom System (A1–A6)

A1 (Local lattice structure). The system is defined on a finite interaction graph $G = (V, E)$, with finite-dimensional Hilbert space $\mathcal{H}_i$ assigned to each vertex $i \in V$. The total Hilbert space is

$$\mathcal{H} = \bigotimes_{i \in V} \mathcal{H}_i.$$

A2 (Local causal update structure). Time evolution is generated by a family of completely positive trace-preserving maps

$$\{\Phi_T\}_{T \geq 0},$$

obtained by composing elementary completely positive trace-preserving updates, each acting nontrivially only on bounded neighborhoods of the interaction graph. There exists a finite interaction radius

$$r < \infty$$

such that every elementary update couples only subsystems whose induced subgraph has graph diameter at most r. Here, locality of an elementary update is understood in the causal sense that an update supported on a region $X \subseteq V$ acts as the identity on the complementary subsystem and cannot enlarge the support of a localized operator or state perturbation beyond the corresponding graph neighborhood determined by the update region.

Moreover, the admissible dynamics possess a uniformly bounded update rate: there exists a finite constant

$$\nu(D) < \infty$$

such that, for every finite evolution time T, the evolution from time 0 to T is generated by at most

$$\nu(D)\,T$$

elementary local updates (equivalently, the update density in physical time is uniformly bounded).

No explicit propagation bound or Lieb–Robinson estimate is assumed. Any finite propagation velocity or causal bound is derived solely from the locality and bounded-rate assumptions above.

A3 (Finite-dimensional bond memory). The dynamics may include auxiliary finite-dimensional memory systems associated with edges $e \in E$. The extended Hilbert space is

$$\mathcal{H}_{\mathrm{ext}} = \mathcal{H} \otimes \mathcal{H}_M, \quad \mathcal{H}_M = \bigotimes_{e \in E} \mathcal{H}_M^{(e)}.$$

A4 (Finite memory mixing). Each memory subsystem evolves under a quantum channel exhibiting exponential mixing. There exist a stationary state σ_M and a finite timescale $\tau_M < \infty$ such that, for every finite-dimensional spectator Hilbert space $\mathcal{H}_S$, the memory mixing estimate remains valid under the natural tensor extension:

$$\|(\mathrm{id}_{\mathcal{H}_S} \otimes \mathcal{T}_M^s)(X) - \mathrm{Tr}_M(X) \otimes \sigma_M\| \leq C e^{-s/\tau_M} \|X\|,$$

for all bounded operators

$$X \in \mathcal{B}(\mathcal{H}_S \otimes \mathcal{H}_M),$$

where Tr_M denotes the partial trace over the memory subsystem. The spectator Hilbert space $\mathcal{H}_S$ represents a mathematical tensor extension used to express stability of the memory mixing property and does not constitute an additional physical subsystem.

A5 (Representable operator subsystems). For each complexity scale D, there exists an operator subsystem

$$\mathcal{A}_D \subseteq \mathcal{B}(\mathcal{H}),$$

representing a class of physically admissible compressed descriptions of the system. No specific representation (e.g., tensor networks) is assumed.

A6 (Admissible dynamics and representability compatibility). For each representability scale D, let

$$\mathfrak{F}_D$$

denote the class of admissible quantum channels. For the operator subsystem

$$\mathcal{A}_D \subseteq \mathcal{B}(\mathcal{H})$$

introduced in Axiom A5, there exists a completely positive unital idempotent map

$$\Pi_D : \mathcal{B}(\mathcal{H}) \longrightarrow \mathcal{A}_D, \qquad \Pi_D^2 = \Pi_D,$$

such that for every admissible evolution

$$\Phi_T \in \mathfrak{F}_D,$$

the compatibility estimate

$$\|\Pi_D \circ \Phi_T \circ \Pi_D(\rho) - \Phi_T \circ \Pi_D(\rho)\| \leq \varepsilon_D(T)$$

holds for every state

$$\rho \in \mathfrak{S}(\mathcal{H}),$$

where

$$\varepsilon_D(T) \longrightarrow 0 \qquad \text{as} \qquad D \longrightarrow \infty$$

for every fixed evolution time T.

Furthermore, for every finite representability scale $D < \infty$, the subsystem

$$\mathcal{A}_D \subsetneq \mathcal{B}(\mathcal{H})$$

is assumed to be a proper unital operator subsystem. Consequently, there exist observables whose expectation values cannot be exactly recovered within $\mathcal{A}_D$, and therefore there exist physically distinguishable quantum states that cannot be distinguished solely through observables contained in $\mathcal{A}_D$. Thus, finite representability constitutes a genuine structural restriction on admissible descriptions, becoming asymptotically complete only in the limit

$$D \to \infty.$$

Norm convention. All norm estimates appearing below are understood with respect to a norm for which the tensor-stable mixing estimate of Axiom A4 holds and which is contractive under the partial trace. In particular,

$$\| \operatorname{Tr}_M(X)\| \leq \|X\|,$$

for every operator X on the corresponding tensor-product Hilbert space. No particular choice of norm is assumed beyond these properties, and no further properties of the norm are used.

Representation Convention

We work primarily in the Schrödinger picture, where

$$\Phi_T : \mathfrak{S}(\mathcal{H}) \to \mathfrak{S}(\mathcal{H})$$

acts on quantum states. When observables are considered, we use the same symbol Φ_T for the dual Heisenberg-picture completely positive unital map

$$\Phi_T : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H}),$$

the intended meaning being determined by context. Likewise, the projection Π_D is defined on observables, and its action on states is understood through the standard Schrödinger–Heisenberg duality.

Admissible dynamics

A quantum channel Φ_T is called admissible if it belongs to $\mathfrak{F}_D$, is induced by a joint evolution on the extended Hilbert space $\mathcal{H}_{\text{ext}}$ of Axiom A3 generated by compositions of local CPTP updates consistent with A2 and A4, and satisfies the representability compatibility condition in A6. The channel Φ_T denotes the resulting reduced evolution on the system Hilbert space $\mathcal{H}$.

3 Entanglement Reachability: Definitions

We now define the notion of dynamical reachability within the axiomatic framework of Section 2.

Definition: Reachable set.

Definition 1. *Let $\mathfrak{S}_D \subseteq \mathfrak{S}(\mathcal{H})$ denote the class of admissible initial states associated with representability scale D.*

The reachable set at time T is defined as

$$\mathcal{R}(T) = \{\rho(T) \mid \rho(0) \in \mathfrak{S}_D,\ \rho(T) = \Phi_T(\rho(0)),\ \Phi_T \in \mathfrak{F}_D\}.$$

The objective of this work is to characterize the structure of $\mathcal{R}(T)$ and determine constraints on entanglement generation under the axioms of Section 2.

For notational simplicity, the dependence of $\mathcal{R}(T)$ on the representability scale D is suppressed throughout when no ambiguity arises.

While the reachable set $\mathcal{R}(T)$ characterizes the states attainable under admissible dynamics, the primary objective of this work is not merely to enumerate reachable states, but to understand the multipartite entanglement patterns that such dynamics can generate. Since many physically distinct states may exhibit equivalent entanglement organization, it is natural to separate the dynamical notion of state reachability from the structural notion of entanglement. We therefore introduce an abstract mapping that associates to each quantum state its multipartite entanglement structure.

Definition: Entanglement-structure map.

Definition 2. *Let*

$$\mathfrak{S}(\mathcal{H})$$

denote the set of density operators on the many-body Hilbert space, and let

$$\mathfrak{E}$$

denote an abstract space of structural configurations; including, in particular, multipartite entanglement structures.

We define the entanglement-structure map

$$\mathcal{E} : \mathfrak{S}(\mathcal{H}) \longrightarrow \mathfrak{E},$$

which is assumed to be a well-defined map. That is, every quantum state

$$\rho \in \mathfrak{S}(\mathcal{H})$$

is assigned a unique multipartite entanglement structure

$$\mathcal{E}(\rho) \in \mathfrak{E}.$$

No particular realization of $\mathfrak{E}$ is assumed. Depending on the physical setting, the image $\mathcal{E}(\rho)$ may be represented through bipartition entanglement, correlation or entanglement graphs, tensor-network connectivity, or other equivalent descriptions of multipartite entanglement. The present framework remains independent of any specific entanglement measure or representation.

We further require that the entanglement-structure map be compatible with admissible dynamics in the following sense: for every admissible evolution

$$\Phi_T \in \mathfrak{F}_D,$$

whenever

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2),$$

one also has

$$\mathcal{E}(\Phi_T(\rho_1)) = \mathcal{E}(\Phi_T(\rho_2))\,.$$

Consequently, every admissible evolution induces a well-defined map on

$$\mathfrak{E},$$

obtained by passing to entanglement structures.

We restrict attention to admissible entanglement-structure maps for which all structural quantities defined on $\mathfrak{E}$ are invariant under the equivalence relation induced by $\mathcal{E}$. Specifically, if

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2),$$

then every induced structural functional satisfies

$$\mathcal{F}(\mathcal{E}(\rho_1)) = \mathcal{F}(\mathcal{E}(\rho_2)).$$

Thus, structural quantities are properties of the structural configuration itself and not of a particular representative state in $\mathfrak{S}(\mathcal{H})$.

The entanglement-structure map introduced above associates to every quantum state an abstract description of its multipartite entanglement organization. Since the dynamics determine only the subset of states that are physically attainable, the corresponding image of the reachable set under $\mathcal{E}$ identifies precisely those entanglement structures that can arise from admissible evolution. Consequently, the study of entanglement reachability naturally reduces to the characterization of the image of $\mathcal{R}(T)$ under the map $\mathcal{E}$.

Accordingly, the central object of this work is not merely the reachable set of quantum states, but the induced set of multipartite entanglement structures generated by admissible dynamics.

Definition: Reachable entanglement structures.

Definition 3. *Let*

$$\mathcal{R}(T) \subseteq \mathfrak{S}(\mathcal{H})$$

denote the reachable set at time T (Definition 3.1), and let

$$\mathcal{E} : \mathfrak{S}(\mathcal{H}) \longrightarrow \mathfrak{E}$$

be the entanglement-structure map (Definition 3.2).

The set of reachable entanglement structures *at time T is defined as the image of the reachable set under $\mathcal{E}$,*

$$\mathfrak{R}_{\mathcal{E}}(T) = \mathcal{E}(\mathcal{R}(T)) = \{\mathcal{E}(\rho) \mid \rho \in \mathcal{R}(T)\} \subseteq \mathfrak{E}.$$

The set $\mathfrak{R}_{\mathcal{E}}(T)$ represents all multipartite entanglement structures that are dynamically attainable under admissible evolutions satisfying the axioms of Section 2. Characterizing the geometry and structural properties of $\mathfrak{R}_{\mathcal{E}}(T)$ constitutes the primary objective of the present work.

The set of reachable entanglement structures introduced above provides a description of the multipartite entanglement structures attainable under admissible dynamics at the level encoded by $\mathcal{E}$. However, individual elements of $\mathfrak{R}_{\mathcal{E}}(T)$ may differ in microscopic realization while exhibiting the same underlying structural organization. For the purpose of understanding the structure of entanglement reachability, it is therefore natural to group structurally equivalent entanglement patterns into common classes rather than treat every reachable structure as a distinct object.

This classification isolates the structural features preserved under an appropriate notion of equivalence and provides a coarse-grained description of the reachable region of entanglement space. The resulting partition forms the basis for the reachability classification developed in the subsequent sections.

The definition of reachable entanglement structures in Definition 3.3 characterizes the subset of multipartite entanglement patterns that arise from dynamically evolved states under admissible evolution. However, this construction is inherently time-dependent and therefore does not exhaust the full representational capacity of the entanglement-structure map $\mathcal{E}$ introduced in Definition 3.2.

In particular, the map $\mathcal{E}$ assigns to *every* quantum state $\rho \in \mathfrak{S}(\mathcal{H})$ an abstract entanglement structure, independently of whether ρ arises from admissible dynamics or from kinematic considerations alone. This suggests a natural distinction between structures that are dynamically realizable within finite time and those that are kinematically well-defined as images of quantum states, irrespective of dynamical accessibility.

To make this distinction precise, it is useful to first isolate the full image of the state space under $\mathcal{E}$, which captures all entanglement structures that are consistent with quantum mechanics at the level of state space alone. This leads to the following definition of kinematically admissible entanglement structures.

Definition: Kinematically admissible entanglement structures.

Definition 4. *The set of* kinematically admissible entanglement structures *is defined as the image of the full state space under the entanglement-structure map:*

$$\mathfrak{E}_{\mathrm{kin}} := \mathcal{E}\big(\mathfrak{S}(\mathcal{H})\big) \subseteq \mathfrak{E}.$$

Equivalently, an entanglement structure $\eta \in \mathfrak{E}$ is kinematically admissible if and only if there exists a quantum state $\rho \in \mathfrak{S}(\mathcal{H})$ such that

$$\eta = \mathcal{E}(\rho).$$

This definition separates purely state-space consistency constraints from dynamical constraints induced by admissible evolutions $\mathfrak{F}_D$.

Structural consistency of kinematic and dynamical entanglement. By construction, the entanglement-structure map $\mathcal{E}$ assigns to every quantum state $\rho \in \mathfrak{S}(\mathcal{H})$ a well-defined element of the ambient entanglement-structure space $\mathfrak{E}$. Since admissible initial states $\mathfrak{S}_D$ are subsets of $\mathfrak{S}(\mathcal{H})$, and admissible dynamics Φ_T act within $\mathfrak{S}(\mathcal{H})$, it follows that all dynamically generated states remain within the domain of $\mathcal{E}$.

Consequently, the set of reachable entanglement structures satisfies

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq \mathfrak{E}_{\mathrm{kin}} \subseteq \mathfrak{E}, \qquad \forall T < \infty,$$

where $\mathfrak{E}_{\mathrm{kin}} := \mathcal{E}(\mathfrak{S}(\mathcal{H}))$ is the full kinematic image of quantum state space under the entanglement-structure map.

This hierarchy distinguishes kinematic admissibility, determined solely by state-space consistency, from dynamical reachability, which is additionally constrained by locality, memory, and representability.

Definition: Structural equivalence relation.

Definition 5. *Let $\mathcal{I} = \{I_\alpha\}$ be a family of structural invariants on the entanglement-structure space $\mathfrak{E}$, where each*

$$I_\alpha : \mathfrak{E} \to \mathcal{X}_\alpha$$

assigns a representation-independent descriptor of multipartite entanglement organization.

We define an equivalence relation $\sim$ on $\mathfrak{E}$ by

$$\eta_1 \sim \eta_2 \quad \Longleftrightarrow \quad I_\alpha(\eta_1) = I_\alpha(\eta_2) \ \ \textit{for all } I_\alpha \in \mathcal{I}.$$

The equivalence classes under $\sim$ define the structural entanglement classes *of the system. All structural invariants and induced structural functionals are understood to be defined on these equivalence classes rather than on individual state representatives.*

Since equality is reflexive, symmetric, and transitive, the relation $\sim$ is an equivalence relation on $\mathfrak{E}$. Consequently, the corresponding quotient

$$\mathfrak{E}/\sim$$

is well defined, and

$$\mathfrak{R}_\mathcal{E}(T)/\sim$$

is its restriction to the reachable subset.

The structural equivalence relation $\sim$ partitions the entanglement-structure space $\mathfrak{E}$ into classes whose members share the same values of the structural invariants $\mathcal{I}$. Restricting this partition to the reachable subset $\mathfrak{R}_\mathcal{E}(T)$ yields a coarse-grained description of the multipartite entanglement structures attainable under admissible dynamics.

The quotient

$$\mathfrak{E}/\sim$$

provides a structural organization of the ambient entanglement-structure space, while its restriction

$$\mathfrak{R}_\mathcal{E}(T)/\sim$$

provides the corresponding coarse-grained organization of the dynamically reachable entanglement structures.

With this organization in place, we now characterize the physically attainable entanglement patterns in terms of these induced equivalence classes.

Definition: Reachable entanglement classes.

Definition 6. *Let*

$$\mathfrak{R}_\mathcal{E}(T) \subseteq \mathfrak{E}$$

denote the set of reachable entanglement structures at time T (Definition 3.3), and let $\sim$ be the structural equivalence relation on $\mathfrak{E}$ defined in terms of the entanglement invariants introduced prior to this definition.

A reachable entanglement class *is the intersection of an equivalence class of $\mathfrak{E}$ under $\sim$ with the reachable set $\mathfrak{R}_\mathcal{E}(T)$, i.e., a subset*

$$\mathcal{C} = [\eta]_\sim \cap \mathfrak{R}_\mathcal{E}(T) \quad \textit{for some } \eta \in \mathfrak{R}_\mathcal{E}(T).$$

The collection of all reachable entanglement classes is given by the restricted quotient

$$\mathfrak{R}_\mathcal{E}(T)/\sim,$$

where this quotient is understood as the restriction of the equivalence relation $\sim$ to the reachable subset. It partitions the set of reachable entanglement structures into disjoint equivalence classes induced by $\sim$.

The equivalence relation $\sim$ is fixed by structural invariants of the entanglement-structure map and does not depend on any particular representation such as bipartition entanglement measures, correlation graphs, or tensor-network realizations, which instead serve only as possible instantiations of those invariants in specific physical settings.

The partition of reachable entanglement structures into equivalence classes provides a structural organization of all multipartite entanglement patterns that can be generated under admissible dynamics. An equally important question is whether every admissible entanglement structure belongs to one of these reachable classes, or whether certain structures remain fundamentally inaccessible.

To formalize this distinction, we introduce the notion of *forbidden entanglement structures.* These correspond to multipartite entanglement organizations that are compatible with the ambient entanglement-structure space but cannot be realized by any admissible evolution from any admissible initial state. The identification of such forbidden structures forms the basis for the reachability constraints established in the subsequent sections.

Definition: Forbidden entanglement structures.

Definition 7. *Let*

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq \mathfrak{E}$$

denote the set of reachable entanglement structures at time T.

An entanglement structure

$$\eta \in \mathfrak{E}$$

is said to be forbidden at time T *if there exists no admissible initial state*

$$\rho(0) \in \mathfrak{S}_D$$

and no admissible evolution

$$\Phi_T \in \mathfrak{F}_D$$

such that

$$\mathcal{E}(\Phi_T(\rho(0))) = \eta.$$

The collection of all forbidden entanglement structures at time T is denoted by

$$\mathfrak{E}_{\text{forb}}(T) = \{\eta \in \mathfrak{E}_{\text{kin}} \mid \eta \notin \mathfrak{R}_{\mathcal{E}}(T)\}.$$

Thus,

$$\mathfrak{E}_{\text{forb}}(T) = \mathfrak{E}_{\text{kin}} \setminus \mathfrak{R}_{\mathcal{E}}(T),$$

where the complement is taken relative to the kinematically admissible multipartite entanglement structures introduced in Definition 4.

Forbidden entanglement structures therefore represent multipartite entanglement organizations that are compatible with the abstract entanglement-structure space but cannot be generated from any admissible initial state under any admissible evolution satisfying the axioms of Section 2.

The notion of forbidden entanglement structures introduced above identifies those multipartite entanglement patterns that cannot be generated under admissible dynamics. However, this distinction implicitly relies on a deeper separation between two distinct notions of admissibility:

one arising purely from the abstract structure of the entanglement-structure space, and another imposed by the constraints of physical evolution.

In particular, not every element of $\mathfrak{E}$ carries the same status with respect to physical realizability. Some elements of the ambient space $\mathfrak{E}$ arise as images of quantum states under the map $\mathcal{E}$, thereby defining the kinematically admissible subset $\mathfrak{E}_{\text{kin}}$, whereas other elements of $\mathfrak{E}$ need not correspond to any quantum state. Among the kinematically admissible structures, only a subset can be generated dynamically under the axioms of Section 2.

This motivates a precise distinction between kinematic admissibility and dynamical reachability, which clarifies the logical structure underlying the classification of reachable, forbidden, and inaccessible entanglement configurations.

Definition: Kinematically admissible and dynamically reachable entanglement structures.

Definition 8. *An entanglement structure*

$$\eta \in \mathfrak{E}$$

is said to be kinematically admissible *if*

$$\eta \in \mathfrak{E}_{\text{kin}},$$

where

$$\mathfrak{E}_{\text{kin}} = \mathcal{E}(\mathfrak{S}(\mathcal{H})) \subseteq \mathfrak{E}$$

is the image of the full quantum state space under the entanglement-structure map (Definition 3.4). Equivalently, η is kinematically admissible if and only if there exists a quantum state

$$\rho \in \mathfrak{S}(\mathcal{H})$$

such that

$$\eta = \mathcal{E}(\rho),$$

independently of whether ρ is dynamically reachable under the admissible evolutions

$$\mathfrak{F}_D.$$

An entanglement structure

$$\eta \in \mathfrak{E}$$

is said to be dynamically reachable at time T *if*

$$\eta \in \mathfrak{R}_{\mathcal{E}}(T),$$

where $\mathfrak{R}_{\mathcal{E}}(T)$ denotes the set of reachable entanglement structures (Definition 3.3).

Equivalently, η is dynamically reachable if there exist an admissible initial state

$$\rho(0) \in \mathfrak{S}_D$$

and an admissible evolution

$$\Phi_T \in \mathfrak{F}_D$$

such that

$$\mathcal{E}(\Phi_T(\rho(0))) = \eta.$$

Accordingly, every dynamically reachable entanglement structure is kinematically admissible, whereas a kinematically admissible structure need not be dynamically reachable. The kinematically admissible but dynamically unreachable structures are precisely the forbidden entanglement structures of Definition 3.5.

The constructions of this section establish a hierarchical reduction of the dynamics-induced state space into a structured space of entanglement descriptors. Starting from the reachable set of quantum states $\mathcal{R}(T)$, we introduced an entanglement-structure map $\mathcal{E}$, thereby inducing the set of reachable entanglement structures $\mathfrak{R}_{\mathcal{E}}(T) \subseteq \mathfrak{E}$. This set was subsequently organized into equivalence classes, allowing for a coarse-grained description of structurally equivalent entanglement configurations. We further identified the complementary notion of forbidden entanglement structures and clarified the distinction between kinematically admissible and dynamically reachable regimes within the entanglement-structure space.

With this framework in place, the problem of entanglement reachability reduces to the analysis of the structure of $\mathfrak{R}_{\mathcal{E}}(T)$, its induced partition into equivalence classes, and the characterization of its complement in $\mathfrak{E}$. In the following section, we develop the main results of this work, providing structural theorems that constrain reachable entanglement classes, identify families of forbidden entanglement structures, and establish rigorous distinctions between kinematic admissibility and dynamical reachability under the axioms of Section 2.

4 Main Results (Theorems on Entanglement Reachability)

Structural restrictions on admissible arguments. All results in this section are formulated without assuming any additional structure on the entanglement-structure space $\mathfrak{E}$ beyond that induced by the entanglement-structure map $\mathcal{E}$ and the invariants $\mathcal{I} = \{I_\alpha\}$ introduced in Section 3. In particular, no metric, topology, or linear structure on $\mathfrak{E}$ is assumed or used unless explicitly derived from these constructions.

All dynamical statements are therefore formulated with respect to admissible evolutions $\Phi_T \in \mathfrak{F}_D$, and all comparisons between entanglement structures are mediated exclusively through the equivalence relation $\sim$.

By the dynamic compatibility condition in Definition 3.2, every admissible evolution

$$\Phi_T \in \mathfrak{F}_D$$

induces a well-defined map on the kinematically admissible structural space,

$$\Gamma_T : \mathfrak{E}_{\text{kin}} \longrightarrow \mathfrak{E}_{\text{kin}},$$

given by

$$\Gamma_T(\mathcal{E}(\rho)) = \mathcal{E}(\Phi_T(\rho)), \qquad \rho \in \mathfrak{S}(\mathcal{H}).$$

The central result of this section is the structural confinement theorem. It establishes that admissible dynamics cannot explore the full entanglement-structure space, but remain confined to a dynamically generated reachable region. All subsequent results, including bounded structural functionals, reachable-region characterizations, non-surjectivity, forbidden-structure results, and the separation between kinematic and dynamical constraints, follow as consequences of this confinement principle.

Induced structural functionals

Defintion: Dynamically induced structural functionals.

Definition 9. *Let $\mathfrak{E}$ be the entanglement-structure space and let $\mathcal{E} : \mathfrak{S}(\mathcal{H}) \to \mathfrak{E}$ be the entanglement-structure map. For each admissible evolution $\Phi_T \in \mathfrak{F}_D$, let*

$$\Gamma_T : \mathfrak{E}_{\text{kin}} \longrightarrow \mathfrak{E}_{\text{kin}}$$

denote the induced map introduced above. Thus, for $\eta = \mathcal{E}(\rho) \in \mathfrak{E}_{\text{kin}}$,

$$\Gamma_T(\eta) = \mathcal{E}(\Phi_T(\rho)).$$

We define a family of induced structural functionals

$$\{\mathcal{F}_\alpha\}_{\alpha\in\mathcal{I}_{\mathrm{str}}}$$

on the kinematically admissible structural space

$$\mathfrak{E}_{\mathrm{kin}} = \mathcal{E}(\mathfrak{S}(\mathcal{H}))$$

by measuring the variation of the corresponding structural invariants under admissible perturbations of the initial state. More precisely, let

$$I_\alpha : \mathfrak{E} \longrightarrow \mathcal{X}_\alpha$$

be a structural invariant from the family introduced in Section 3, where each codomain

$$\mathcal{X}_\alpha$$

is equipped with a prescribed comparison map

$$d_\alpha : \mathcal{X}_\alpha \times \mathcal{X}_\alpha \longrightarrow [0,\infty),$$

used only to compare values of the invariant.

Then

$$\mathcal{F}_\alpha(\eta) := \sup_{\rho(\cdot)\in\mathcal{T}_\alpha(\eta,\rho)} d_\alpha\big(I_\alpha\big(\Gamma_T(\mathcal{E}(\rho(1)))\big), I_\alpha\big(\Gamma_T(\mathcal{E}(\rho(0)))\big)\big)\,,$$

where

$$\eta = \mathcal{E}(\rho(0)),$$

and

$$\mathcal{T}_\alpha(\eta,\rho)$$

denotes a class of admissible state perturbation paths

$$\rho(s) \in \mathfrak{S}(\mathcal{H}), \qquad s \in [0,1],$$

with

$$\rho(0) = \rho.$$

Each perturbation path

$$\rho(s) \in \mathcal{T}_\alpha(\eta,\rho)$$

is assumed to be generated by a local admissible perturbation supported on a finite vertex set

$$S \subseteq V.$$

That is, there exists a family of local completely positive trace-preserving maps

$$\Lambda_s^{(S)}$$

acting nontrivially only on the subsystem associated with S such that

$$\rho(s) = \Lambda_s^{(S)}(\rho).$$

The minimal support of the perturbation is defined as the smallest subset

$$S \subseteq V$$

with this property and is denoted by

$$\mathrm{supp}(\Lambda^{(S)}) = S.$$

Consistent with the admissibility convention for structural maps introduced in Section 3, the perturbation family $\mathcal{T}_\alpha(\eta,\rho)$ is chosen so that whenever

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2) = \eta,$$

the corresponding suprema agree:

$$\sup_{\delta\rho\in\mathcal{T}_\alpha(\eta,\rho_1)} d_\alpha\big(I_\alpha(\Gamma_T(\mathcal{E}(\rho_1+\delta\rho))), I_\alpha(\Gamma_T(\eta))\big) = \sup_{\delta\rho\in\mathcal{T}_\alpha(\eta,\rho_2)} d_\alpha\big(I_\alpha(\Gamma_T(\mathcal{E}(\rho_2+\delta\rho))), I_\alpha(\Gamma_T(\eta))\big)\,.$$

Consequently, the value of

$$\mathcal{F}_\alpha(\eta)$$

is independent of the choice of representative

$$\rho \in \mathfrak{S}(\mathcal{H})$$

satisfying

$$\mathcal{E}(\rho) = \eta,$$

and therefore $\mathcal{F}_\alpha$ is a well-defined functional on

$$\mathfrak{E}_{\mathrm{kin}}.$$

The definition depends only on the structural invariants introduced in Section 3 and therefore introduces no additional metric, topological, or linear structure on the entanglement-structure space $\mathfrak{E}$ itself.

Lemma: Memory-induced contraction of reduced system states.

Lemma 1. *Let*

$$\widehat{\mathcal{T}}_M := \mathrm{id}_{\mathcal{H}} \otimes \mathcal{T}_M$$

denote the extension of the memory channel of Axiom A4 to the extended Hilbert space

$$\mathcal{H}_{\mathrm{ext}} = \mathcal{H} \otimes \mathcal{H}_M.$$

Then for any two extended density operators

$$\rho_1, \rho_2 \in \mathfrak{S}(\mathcal{H}_{\mathrm{ext}})$$

satisfying

$$\mathrm{Tr}_M(\rho_1) = \mathrm{Tr}_M(\rho_2),$$

the reduced system states satisfy

$$\Big\|\mathrm{Tr}_M\big(\widehat{\mathcal{T}}_M^s(\rho_1)\big) - \mathrm{Tr}_M\big(\widehat{\mathcal{T}}_M^s(\rho_2)\big)\Big\| \le Ce^{-s/\tau_M}\|\rho_1 - \rho_2\|.$$

The preceding estimate shows that differences between extended states with identical system marginals become exponentially suppressed after reduction to the system degrees of freedom. This operator-level contraction serves as the dynamical input for the structural estimates developed later through the induced structural functionals of Definition 9.

Proof. Let

$$X := \rho_1 - \rho_2.$$

Since

$$\mathrm{Tr}_M(\rho_1) = \mathrm{Tr}_M(\rho_2),$$

we have

$$\mathrm{Tr}_M(X) = 0.$$

By Axiom A4 in its tensor-stable form, applied to the extended system Hilbert space

$$\mathcal{H}_S = \mathcal{H},$$

the operator X satisfies

$$\|(\mathrm{id}_{\mathcal{H}} \otimes \mathcal{T}_M^s)(X) - \mathrm{Tr}_M(X) \otimes \sigma_M\| \leq C e^{-s/\tau_M} \|X\|.$$

Because

$$\mathrm{Tr}_M(X) = 0,$$

and since

$$\widehat{\mathcal{T}}_M^s = \mathrm{id}_{\mathcal{H}} \otimes \mathcal{T}_M^s,$$

this reduces to

$$\left\| \widehat{\mathcal{T}}_M^s(X) \right\| \leq C e^{-s/\tau_M} \|X\|.$$

Using the contractivity of the partial trace with respect to the norm appearing in Axiom A4, we obtain

$$\left\| \mathrm{Tr}_M \left(\widehat{\mathcal{T}}_M^s(X) \right) \right\| \leq C e^{-s/\tau_M} \|X\|.$$

Substituting

$$X = \rho_1 - \rho_2$$

gives

$$\left\| \mathrm{Tr}_M \big(\widehat{\mathcal{T}}_M^s(\rho_1) \big) - \mathrm{Tr}_M \big(\widehat{\mathcal{T}}_M^s(\rho_2) \big) \right\| \leq C e^{-s/\tau_M} \|\rho_1 - \rho_2\|,$$

which proves the claim.

□

Lemma: Locality-induced bounded propagation of perturbations.

Lemma 2. *Assume Axioms A1–A2.*

Let

$$\Phi_T \in \mathfrak{F}_D$$

be an admissible evolution, extended linearly from states to the trace-class operator space

$$\mathcal{T}(\mathcal{H}).$$

Let

$$\delta\rho \in \mathcal{T}(\mathcal{H})$$

be a localized perturbation satisfying

$$\mathrm{supp}(\delta\rho) \subseteq S,$$

for a finite vertex set

$$S \subseteq V.$$

Then there exists a finite effective propagation velocity

$$v_{\mathrm{eff}}(D) < \infty,$$

depending only on the locality parameters of the admissible dynamicsdepending only on the interaction radius r and the update-rate bound $\nu(D)$ from Axiom A2, such that for every finite evolution time T,

$$\mathrm{supp}\,(\Phi_T(\delta\rho)) \subseteq B_{v_{\mathrm{eff}}(D)T}(S),$$

where

$$B_R(S) = \{x \in V : \mathrm{dist}(x, S) \le R\}$$

denotes the graph-metric neighbourhood of radius R.

Consequently, if

$$\Gamma_T = \mathcal{E} \circ \Phi_T$$

is the induced entanglement-structure evolution, then the evolved perturbation

$$\Phi_T(\delta\rho)$$

entering the construction of

$$\Gamma_T(\mathcal{E}(\rho + \delta\rho))$$

is supported within the locality region

$$B_{v_{\mathrm{eff}}(D)T}(S).$$

Hence, locality provides the geometric constraint on perturbations that enter the subsequent construction of the induced structural functionals

$$\{\mathcal{F}_\alpha\}_{\alpha \in \mathcal{I}_{\mathrm{str}}}.$$

Proof. By Axiom A2, every elementary update acts only on a subgraph of graph diameter at most r. Moreover, Axiom A2 asserts the existence of a uniform bound

$$\nu(D) < \infty$$

on the number of elementary updates per unit physical time.

Let

$$\delta\rho$$

be supported on the finite vertex set

$$S.$$

By Axiom A2, each elementary update acts nontrivially only within a subgraph of diameter at most r. Therefore, under the Schrödinger-picture action of an elementary update, the support of a localized perturbation can enlarge by at most a graph distance r. Since no more than

$$\nu(D)T$$

elementary updates occur during the time interval $[0, T]$, the support of the propagated perturbation can expand by at most

$$r\,\nu(D)\,T.$$

Defining

$$v_{\mathrm{eff}}(D) := r\,\nu(D),$$

we obtain

$$\mathrm{supp}(\Phi_T(\delta\rho)) \subseteq B_{v_{\mathrm{eff}}(D)T}(S).$$

Now consider the induced evolution

$$\Gamma_T = \mathcal{E} \circ \Phi_T.$$

Since Γ_T depends only on the evolved quantum state, the effect of the initial perturbation on the resulting structural configuration is mediated entirely through the evolved state difference

$$\Phi_T(\rho + \delta\rho) - \Phi_T(\rho) = \Phi_T(\delta\rho).$$

The preceding support estimate therefore provides a locality constraint on the quantum-state variation entering the construction of the induced structural functionals

$$\{\mathcal{F}_\alpha\}_{\alpha \in \mathcal{I}_{\mathrm{str}}},$$

without requiring any additional locality assumption on the structural map $\mathcal{E}$. Thus every perturbation entering the definition of the induced structural functionals is confined to this bounded locality region under admissible evolution, establishing the asserted geometric confinement.

$\square$

Lemma: Representability compatibility and approximate subsystem invariance.

Lemma 3. *Let*

$$\Pi_D : \mathcal{B}(\mathcal{H}) \to \mathcal{A}_D$$

be the completely positive unital idempotent map introduced in Axiom A6, and let

$$\Phi_T \in \mathfrak{F}_D$$

be an admissible evolution.

Then the following statements hold.

(i) For every state

$$\rho \in \mathfrak{S}(\mathcal{H}),$$

the compatibility estimate

$$\|\Pi_D \circ \Phi_T \circ \Pi_D(\rho) - \Phi_T \circ \Pi_D(\rho)\| \le \varepsilon_D(T)$$

holds, where

$$\varepsilon_D(T) \longrightarrow 0 \qquad \textit{as} \qquad D \longrightarrow \infty.$$

Consequently, admissible dynamics remain compatible with the representable subsystem $\mathcal{A}_D$ up to the controlled error

$$\varepsilon_D(T).$$

*(ii) (*Approximate invariance of the representable subsystem.*) For every finite representability scale*

$$D < \infty,$$

the dual Heisenberg-picture evolution maps the representable operator subsystem to within $\varepsilon_D(T)$ of itself, in the sense that

$$\Phi_T(\mathcal{A}_D) \subseteq_{\varepsilon_D(T)} \mathcal{A}_D,$$

where $\subseteq_{\varepsilon_D(T)}$ denotes inclusion up to an error bounded by $\varepsilon_D(T)$.

(iii) Since

$$\mathcal{A}_D \subsetneq \mathcal{B}(\mathcal{H})$$

for every finite representability scale

$$D < \infty,$$

the representability projection

$$\Pi_D$$

is nontrivial.

Consequently, there exist operators in

$$\mathcal{B}(\mathcal{H})$$

whose information is not retained under projection onto

$$\mathcal{A}_D.$$

Accordingly, representability compatibility imposes a genuine restriction on the operator information preserved by admissible evolution at finite representability scale.

Proof. Statement (i) is precisely the compatibility condition of Axiom A6.

To prove (ii), let

$$X \in \mathcal{A}_D.$$

Since

$$\Pi_D$$

is idempotent,

$$\Pi_D(X) = X.$$

Applying the compatibility estimate of Axiom A6 to X yields

$$\|\Pi_D(\Phi_T(X)) - \Phi_T(X)\| \leq \varepsilon_D(T).$$

Since

$$\Pi_D(\Phi_T(X)) \in \mathcal{A}_D,$$

the evolved operator

$$\Phi_T(X)$$

lies within a distance at most $\varepsilon_D(T)$ of the representable subsystem. Hence

$$\Phi_T(\mathcal{A}_D) \subseteq_{\varepsilon_D(T)} \mathcal{A}_D,$$

establishing the asserted approximate invariance.

Finally, consider statement (iii).

Since

$$\mathcal{A}_D \subsetneq \mathcal{B}(\mathcal{H}),$$

there exists an operator

$$X \in \mathcal{B}(\mathcal{H}) \setminus \mathcal{A}_D.$$

Because the quantum state space separates observables, there exist states

$$\rho_1, \rho_2 \in \mathfrak{S}(\mathcal{H})$$

such that

$$\mathrm{Tr}[(\rho_1 - \rho_2)X] \neq 0.$$

Therefore no collection of expectation values of observables contained entirely within

$$\mathcal{A}_D$$

can, in general, determine the expectation value of X.

Hence finite representability removes operator information that is unavailable within the representable subsystem, establishing that Axiom A6 provides an independent structural restriction on admissible dynamics.

□

Definition: Admissible structural observables and induced structural functionals.

Definition 10. *Let*

$$\mathcal{E} : \mathfrak{S}(\mathcal{H}) \longrightarrow \mathfrak{E}$$

be the entanglement-structure map.

An admissible structural observable is a real-valued functional

$$O : \mathfrak{S}(\mathcal{H}) \longrightarrow \mathbb{R},$$

interpreted as a structural quantity evaluated on quantum states rather than as an observable in the operator-algebraic sense. It is required to be constant on the fibres of the entanglement-structure map, namely,

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2) \quad \Longrightarrow \quad O(\rho_1) = O(\rho_2).$$

The collection of all admissible structural observables is denoted by

$$\mathcal{O}_{\rm str}.$$

An admissible structural observable

$$O \in \mathcal{O}_{\rm str}$$

is furthermore required to satisfy the following compatibility conditions.

- *(a)* (Locality compatibility). *The value of O depends only on structural information that is accessible under admissible evolutions belonging to $\mathfrak{F}_D$, and is insensitive to perturbations excluded by the locality constraints of Axiom A2.*
- *(b)* (Memory stability). *Along every admissible evolution, the value of O is stable under the asymptotically mixing memory dynamics prescribed by Axiom A4, up to the decay estimates established there.*
- *(c)* (Representability compatibility). *The value of O depends only on information preserved under the representability projection Π_D, up to the approximation error $\varepsilon_D(T)$ appearing in Axiom A6.*

Let

$$C_O(T, D)$$

denote the uniform bound on the induced structural functional established for admissible evolutions in Proposition 1.

The class

$$\mathcal{O}_{\rm str}$$

is furthermore assumed to be dynamically complete in the sense of structural detection*. Namely, for every finite evolution time*

$$T < \infty,$$

and every entanglement structure

$$\eta^\star \in \mathfrak{E} \setminus \mathfrak{R}_{\mathcal{E}}(T),$$

there exists an admissible structural observable

$$O \in \mathcal{O}_{\rm str}$$

such that

$$\mathcal{F}_O(\eta^\star) > C_O(T, D).$$

Thus the family of induced structural functionals is assumed to detect all dynamically inaccessible structural features through the corresponding boundedness thresholds.

In addition, every admissible structural observable

$$O \in \mathcal{O}_{\mathrm{str}}$$

is assumed to be uniformly bounded on the entanglement-structure classes. Equivalently, there exists a constant

$$M_O < \infty$$

such that

$$|O(\rho)| \leq M_O$$

for every

$$\rho \in \mathfrak{S}(\mathcal{H}).$$

Every

$$O \in \mathcal{O}_{\mathrm{str}}$$

therefore induces a structural functional

$$\mathcal{F}_O : \mathfrak{E} \longrightarrow \mathbb{R}$$

defined by

$$\mathcal{F}_O(\eta) := O(\rho), \qquad \mathcal{E}(\rho) = \eta,$$

which is well defined, since any two representatives $\rho_1, \rho_2 \in \mathfrak{S}(\mathcal{H})$ satisfying $\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2) = \eta$ have $O(\rho_1) = O(\rho_2)$ by construction.

Proposition: Construction of induced structural functionals

Proposition 1. *Let*

$$\Gamma_T = \mathcal{E} \circ \Phi_T : \mathfrak{S}_D \longrightarrow \mathfrak{E}$$

be the induced entanglement-structure evolution associated with an admissible evolution $\Phi_T \in \mathfrak{F}_D$ satisfying Axioms A1–A6.

Let

$$\mathcal{O}_{\mathrm{str}}$$

denote the class of admissible structural observables introduced in Definition 10. For every

$$O \in \mathcal{O}_{\mathrm{str}},$$

let

$$\mathcal{F}_O : \mathfrak{E} \to \mathbb{R}$$

denote the induced structural functional introduced in Definition 10.

For

$$O \in \mathcal{O}_{\mathrm{str}},$$

define

$$C_O(T, D) := \sup_{\xi \in \Gamma_T(\mathfrak{R}_{\mathcal{E}}(0))} \mathcal{F}_O(\xi).$$

Then $C_O(T, D)$ is finite and satisfies

$$\mathcal{F}_O(\Gamma_T(\eta)) \leq C_O(T, D)$$

for every

$$\eta \in \mathfrak{R}_{\mathcal{E}}(0).$$

Consequently, every admissible structural observable induces a uniformly bounded structural functional along admissible evolutions on $\mathfrak{E}$, and the corresponding family of induced structural functionals provides uniformly bounded structural quantities on every reachable entanglement structure.

Proof. The proof proceeds in four steps.

Step 1. Well-definedness of the induced structural functional.
Let

$$O \in \mathcal{O}_{\mathrm{str}}$$

be an admissible structural observable, and let

$$\mathcal{F}_O$$

be defined as in Definition 10. Suppose

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2) = \eta.$$

By Definition 10, O is constant on the fibres of the entanglement-structure map, so

$$O(\rho_1) = O(\rho_2).$$

Hence

$$\mathcal{F}_O(\eta) = O(\rho)$$

is independent of the chosen representative ρ, and therefore

$$\mathcal{F}_O : \mathfrak{E} \longrightarrow \mathbb{R}$$

is a well-defined functional.

Step 2. Locality compatibility.
Let

$$\Gamma_T = \mathcal{E} \circ \Phi_T, \qquad \Phi_T \in \mathfrak{F}_D.$$

By Lemma 2, every localized perturbation propagates only inside the bounded neighbourhood

$$B_{v_{\mathrm{eff}}(D)T}(S),$$

where S denotes the initial support of the perturbation. Since O satisfies the locality compatibility condition of Definition 10, its value depends only on structural information generated by admissible local evolutions. Consequently, the induced functional

$$\mathcal{F}_O$$

depends only on structural information accessible within the locality region determined by Lemma 2.

Step 3. Stability under memory mixing.
By Lemma 1, the auxiliary memory dynamics contract exponentially toward the stationary memory state. Moreover, Definition 10 requires every admissible structural observable to satisfy the memory-stability condition. Hence the value of

$$O$$

along admissible evolutions remains uniformly controlled by the decay estimates established in Lemma 1, so no unbounded contribution can arise from persistent memory effects.

Step 4. Representability compatibility and boundedness.

By Lemma 3, admissible evolutions remain compatible with the representable subsystem

$$\mathcal{A}_D$$

up to the approximation error

$$\varepsilon_D(T).$$

Since every admissible structural observable is representability compatible by Definition 10, its value is determined solely by information retained within the representable subsystem up to the same controlled error.

By Definition 10, every admissible structural observable is uniformly bounded. Hence there exists

$$M_O < \infty$$

such that, for every

$$\eta \in \mathfrak{E},$$

and every representative

$$\rho \in \mathfrak{S}(\mathcal{H})$$

satisfying

$$\mathcal{E}(\rho) = \eta,$$

one has

$$|\mathcal{F}_O(\eta)| = |O(\rho)| \leq M_O.$$

In particular,

$$C_O(T, D) := \sup_{\xi \in \Gamma_T(\mathfrak{R}_{\mathcal{E}}(0))} \mathcal{F}_O(\xi)$$

is finite and satisfies

$$C_O(T, D) \leq M_O.$$

Moreover,

$$\mathcal{F}_O(\Gamma_T(\eta)) \leq C_O(T, D)$$

for every

$$\eta \in \mathfrak{R}_{\mathcal{E}}(0).$$

Thus every admissible structural observable induces a well-defined structural functional on the entanglement-structure space that is uniformly bounded. In particular, its restriction to every admissible evolution remains uniformly bounded.

□

Hypothesis (Dynamic completeness of admissible structural observables). The class

$$\mathcal{O}_{\mathrm{str}}$$

is assumed to be *dynamically complete in the sense of structural detection.* Namely, for every finite evolution time

$$T < \infty,$$

and every entanglement structure

$$\eta^\star \in \mathfrak{E} \setminus \mathfrak{R}_{\mathcal{E}}(T),$$

there exists an admissible structural observable

$$O \in \mathcal{O}_{\mathrm{str}}$$

such that

$$\mathcal{F}_O(\eta^\star) > C_O(T, D).$$

Thus the family of induced structural functionals is assumed to detect all dynamically inaccessible structural features through the corresponding boundedness thresholds.

Role of the dynamic completeness hypothesis. The preceding hypothesis is a structural completeness hypothesis rather than a consequence of Axioms A1–A6. Its purpose is to assert that the family of admissible structural observables is sufficiently rich to separate every dynamically inaccessible entanglement structure from the reachable region through at least one induced structural functional. Accordingly, the role of the hypothesis is not to generate the structural confinement itself, but to ensure that the induced structural functionals provide an exhaustive description of that confinement. The logical organization of the theory is therefore summarized by

$$\boxed{\begin{aligned}\text{Axioms A1–A6} &\Longrightarrow \text{structural confinement}\\ &\Longrightarrow \text{admissible structural observables encode the confinement}\\ &\Longrightarrow \text{dynamic completeness asserts that this encoding is exhaustive.}\end{aligned}}$$

Thus the dynamic completeness hypothesis plays a role analogous to completeness or separating-family hypotheses that occur throughout functional analysis, topology, and resource theories: the underlying structural constraints are established independently, while completeness guarantees that the chosen family of observables separates every dynamically forbidden entanglement structure.

Theorem: Structural confinement of reachable entanglement structures.

Theorem 1. *Let*

$$\mathcal{R}(T) \subseteq \mathfrak{S}(\mathcal{H})$$

denote the reachable set of states at time T under admissible dynamics satisfying Axioms A1–A6, and let

$$\mathcal{E} : \mathfrak{S}(\mathcal{H}) \longrightarrow \mathfrak{E}$$

be the entanglement-structure map.

Let

$$\{\mathcal{F}_O\}_{O\in\mathcal{O}_{\mathrm{str}}}$$

be the family of induced structural functionals obtained in Proposition 1.

Then, for every reachable state

$$\rho(T) \in \mathcal{R}(T),$$

the corresponding entanglement structure satisfies

$$\mathcal{F}_O(\mathcal{E}(\rho(T))) \leq C_O(T, D), \qquad \forall\, O \in \mathcal{O}_{\mathrm{str}},$$

where each bound $C_O(T, D)$ depends on the evolution time T, the representability scale D, the fixed parameters of the admissible dynamics, and the corresponding reachable set generated by the admissible evolution, while remaining independent of the microscopic realization of the initial state.

Consequently, the reachable entanglement structures satisfy the structural confinement

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq \bigcap_{O\in\mathcal{O}_{\mathrm{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \leq C_O(T, D)\}\,.$$

In particular, the family $\{\mathcal{F}_O\}_{O\in\mathcal{O}_{\mathrm{str}}}$ jointly characterizes the admissible confinement region of $\mathfrak{R}_{\mathcal{E}}(T)$ within the ambient entanglement-structure space $\mathfrak{E}$. The theorem shows that, in addition to locality compatibility, the memory stability condition of Axiom A4 and the representability compatibility constraints encoded in Axiom A6 contribute to the structural confinement of reachable entanglement structures through the induced functionals

$$\{\mathcal{F}_O\}_{O\in\mathcal{O}_{\mathrm{str}}}.$$

Thus, the reachable entanglement structures occupy a dynamically defined structural region within the locality-compatible entanglement-structure space. The properness of this confinement region is established in Theorem 2.

Proof. Let

$$\rho(T) \in \mathcal{R}(T)$$

be an arbitrary reachable state, and let

$$\eta = \mathcal{E}(\rho(T))$$

denote its associated entanglement structure.

Since

$$\rho(T) \in \mathcal{R}(T),$$

there exists an admissible evolution producing $\rho(T)$. Applying Proposition 1 to the corresponding admissible evolution yields, for every

$$O \in \mathcal{O}_{\mathrm{str}},$$

the estimate

$$\mathcal{F}_O(\eta) \leq C_O(T, D),$$

for the induced structural functional. The constant

$$C_O(T, D)$$

depends on the evolution time, the representability scale, the fixed parameters of the admissible dynamics, and the corresponding reachable set generated by the admissible evolution. In particular, it is independent of the microscopic realization of the initial state.

Since the above estimate holds for every admissible structural observable, the entanglement structure

$$\eta$$

belongs to every sublevel set generated by an admissible structural observable

$$\left\{\eta' \in \mathfrak{E} \;\middle|\; \mathcal{F}_O(\eta') \leq C_O(T, D)\right\}.$$

Therefore,

$$\eta \in \bigcap_{O \in \mathcal{O}_{\mathrm{str}}} \left\{\eta' \in \mathfrak{E} \;\middle|\; \mathcal{F}_O(\eta') \leq C_O(T, D)\right\}.$$

Since

$$\rho(T) \in \mathcal{R}(T)$$

was arbitrary, it follows that

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq \bigcap_{O \in \mathcal{O}_{\mathrm{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \leq C_O(T, D)\},$$

which proves the asserted structural confinement of $\mathfrak{R}_{\mathcal{E}}(T)$.

This completes the proof. □

Proposition: Properness of the structural confinement region.

Proposition 2. *Let*

$$\Gamma_T = \mathcal{E} \circ \Phi_T : \mathfrak{S}_D \longrightarrow \mathfrak{E}$$

be the induced entanglement-structure evolution under admissible dynamics satisfying Axioms A1–A6.

Then, for every finite evolution time

$$T < \infty,$$

the structural confinement region obtained in Theorem 1,

$$\bigcap_{O \in \mathcal{O}_{\mathrm{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \leq C_O(T, D)\},$$

is a proper subset of the kinematically admissible entanglement-structure space:

$$\bigcap_{O \in \mathcal{O}_{\mathrm{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \leq C_O(T, D)\} \subsetneq \mathfrak{E}.$$

Consequently, the induced evolution is not surjective:

$$\Gamma_T(\mathfrak{S}_D) = \mathfrak{R}_{\mathcal{E}}(T) \subsetneq \mathfrak{E}.$$

Proof. Assume, for contradiction, that

$$\Gamma_T : \mathfrak{S}_D \longrightarrow \mathfrak{E}$$

is surjective. Then

$$\Gamma_T(\mathfrak{S}_D) = \mathfrak{E},$$

and therefore

$$\mathfrak{R}_{\mathcal{E}}(T) = \mathfrak{E}.$$

However, Theorem 1 implies that every reachable entanglement structure satisfies

$$\mathcal{F}_O(\eta) \leq C_O(T, D)$$

for every

$$O \in \mathcal{O}_{\mathrm{str}}.$$

Hence Theorem 1 yields

$$\mathfrak{E} = \mathfrak{R}_{\mathcal{E}}(T) \subseteq \bigcap_{O \in \mathcal{O}_{\mathrm{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \leq C_O(T, D)\}.$$

Since the intersection on the right is, by construction, a subset of $\mathfrak{E}$, it follows that

$$\bigcap_{O \in \mathcal{O}_{\mathrm{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \leq C_O(T, D)\} = \mathfrak{E}.$$

In other words, the confinement region would coincide with the entire entanglement-structure space.

But by the Dynamic Completeness Hypothesis, there exists a kinematically admissible entanglement structure

$$\eta^\star \in \mathfrak{E},$$

together with an admissible structural observable

$$O \in \mathcal{O}_{\text{str}},$$

such that

$$\mathcal{F}_O(\eta^\star) > C_O(T, D).$$

Thus

$$\eta^\star \notin \bigcap_{O \in \mathcal{O}_{\text{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \le C_O(T, D)\},$$

contradicting the conclusion that this intersection equals $\mathfrak{E}$.

Hence the confinement region is proper:

$$\bigcap_{O \in \mathcal{O}_{\text{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \le C_O(T, D)\} \subsetneq \mathfrak{E}.$$

Finally, Theorem 1 establishes that

$$\mathfrak{R}_{\mathcal{E}}(T)$$

is contained in the above proper subset of $\mathfrak{E}$. Hence

$$\mathfrak{R}_{\mathcal{E}}(T) \subsetneq \mathfrak{E}.$$

Since

$$\Gamma_T(\mathfrak{S}_D) = \mathfrak{R}_{\mathcal{E}}(T),$$

the induced evolution

$$\Gamma_T : \mathfrak{S}_D \longrightarrow \mathfrak{E}$$

cannot be surjective for any finite evolution time, which completes the proof.

□

Theorem: Existence of dynamically forbidden entanglement families.

Theorem 2. *Let*

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq \mathfrak{E}$$

denote the reachable entanglement structures under admissible dynamics satisfying Axioms A1–A6.

Then, for every finite evolution time

$$T < \infty,$$

there exists at least one kinematically admissible entanglement structure that is not reachable by any admissible evolution. Equivalently,

$$\mathfrak{E}_{\text{forb}}(T) \neq \varnothing.$$

Equivalently, there exists a nonempty family of kinematically admissible entanglement structures

$$\{\eta^\star_\lambda\}_{\lambda \in \Lambda} \subseteq \mathfrak{E}$$

such that

$$\eta^\star_\lambda \notin \mathfrak{R}_{\mathcal{E}}(T), \qquad \forall \lambda \in \Lambda.$$

Moreover, these forbidden structures are dynamically characterized by the failure of at least one structural confinement inequality. Namely, for every

$$\eta^\star_\lambda \in \mathfrak{E}_{\text{forb}}(T),$$

there exists

$$O \in \mathcal{O}_{\mathrm{str}}$$

such that

$$\mathcal{F}_O(\eta_\lambda^\star) > C_O(T, D).$$

Hence the forbidden family is not merely defined as the set-theoretic complement of the reachable region; rather, it is detected by the induced structural functionals associated with admissible structural observables.

Proof. By Proposition 2, the induced entanglement-structure evolution

$$\Gamma_T = \mathcal{E} \circ \Phi_T : \mathfrak{S}_D \longrightarrow \mathfrak{E}$$

fails to be surjective for every finite evolution time

$$T < \infty.$$

Therefore,

$$\Gamma_T(\mathfrak{S}_D) \neq \mathfrak{E}.$$

By definition of the reachable entanglement-structure set,

$$\Gamma_T(\mathfrak{S}_D) = \mathfrak{R}_{\mathcal{E}}(T),$$

and hence

$$\mathfrak{R}_{\mathcal{E}}(T) \subsetneq \mathfrak{E}.$$

This proves that the admissible dynamics generate only a proper subset of the kinematically admissible entanglement-structure space.

Consequently, the complement

$$\mathfrak{E}_{\mathrm{forb}}(T) := \mathfrak{E} \setminus \mathfrak{R}_{\mathcal{E}}(T)$$

is nonempty. Therefore there exists a nonempty family

$$\{\eta_\lambda^\star\}_{\lambda \in \Lambda} \subseteq \mathfrak{E}$$

such that

$$\eta_\lambda^\star \notin \mathfrak{R}_{\mathcal{E}}(T), \qquad \forall \lambda \in \Lambda.$$

It remains to show that this exclusion is dynamically characterized rather than merely defined as a set-theoretic complement. By the dynamical completeness assumption in Definition 10, any structure outside the reachable region violates at least one structural confinement bound. Hence, for every

$$\eta^\star \in \mathfrak{E}_{\mathrm{forb}}(T),$$

there exists an admissible structural observable

$$O \in \mathcal{O}_{\mathrm{str}}$$

such that

$$\mathcal{F}_O(\eta^\star) > C_O(T, D).$$

Therefore the forbidden family is detected by the induced structural functionals arising from admissible dynamics. The excluded structures are not merely absent from the image of Γ_T, but are separated from the reachable region by the dynamical confinement bounds established in Theorem 1.

This completes the proof. □

Interpretation of Theorem 2. Theorem 2 should not be interpreted merely as an existence result asserting that dynamically forbidden entanglement structures must occur. Rather, it provides a universal construction principle for identifying forbidden structural regions within any admissible realization of the framework.

Once a structural state space, an entanglement-structure space, an admissible representable operator subsystem satisfying Axioms A5–A6, and a family of admissible structural observables have been specified, the theorem automatically associates structural confinement inequalities with those observables. The corresponding violation sets determine dynamically forbidden regions of the structural space.

Consequently, the theorem is independent of any particular physical realization. Different realizations need not produce the same confinement inequality or exclude the same family of structures. Instead, each admissible realization inherits its own confinement criterion from the common abstract framework.

The explicit realizations developed in Section 5 should therefore be understood as concrete instantiations of this universal construction principle rather than as independent no-go results. Their role is to demonstrate how different choices of structural observables generate different forbidden regions while remaining governed by the same abstract confinement mechanism.

Corollary: Dynamical reachability is strictly stronger than kinematic admissibility.

Corollary 1. *For every finite evolution time*

$$T < \infty,$$

kinematic admissibility is not sufficient for dynamical reachability. Equivalently, there exist kinematically admissible entanglement structures that cannot be generated by any admissible evolution in finite time.

Thus dynamical reachability defines a strictly stronger notion than kinematic admissibility for entanglement structures.

Proof. By Theorem 2, the forbidden structural set

$$\mathfrak{E}_{\text{forb}}(T) = \mathfrak{E} \setminus \mathfrak{R}_{\mathcal{E}}(T)$$

is nonempty for every finite evolution time. Hence there exist kinematically admissible entanglement structures that do not belong to the reachable set

$$\mathfrak{R}_{\mathcal{E}}(T).$$

Therefore, not every kinematically admissible entanglement structure is dynamically reachable. Consequently, dynamical reachability is a strictly stronger notion than kinematic admissibility.

□

Theorem: Canonical structural confinement hull.

Theorem 3. *For every finite evolution time*

$$T < \infty,$$

define the structural confinement hull

$$\mathfrak{C}(T) := \bigcap_{O \in \mathcal{O}_{\text{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \leq C_O(T, D)\}.$$

Then the following hold.

(i) The reachable entanglement structures satisfy

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq \mathfrak{C}(T).$$

(ii) The structural confinement hull is a proper subset of the kinematically admissible entanglement-structure space:

$$\mathfrak{C}(T) \subsetneq \mathfrak{E}.$$

(iii) The structural confinement hull is the minimal structural confinement envelope generated by the admissible structural functionals through structural inequalities. Precisely, if

$$X = \bigcap_{O \in \mathcal{O}_{\mathrm{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \leq B_O\},$$

for some family of constants

$$B_O \geq C_O(T, D),$$

and

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq X,$$

then

$$\mathfrak{C}(T) \subseteq X.$$

Thus

$$\mathfrak{C}(T)$$

is the unique minimal structural confinement envelope generated by the admissible structural functionals and containing the dynamically reachable entanglement structures.

Proof. Part (i) is precisely the content of Theorem 1, which establishes that every reachable entanglement structure satisfies all structural confinement inequalities. Hence

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq \mathfrak{C}(T).$$

Part (ii) follows directly from Proposition 2, which proves that the intersection of all admissible structural confinement inequalities is a proper subset of $\mathfrak{E}$. Therefore,

$$\mathfrak{C}(T) \subsetneq \mathfrak{E}.$$

Part (iii). It remains to establish the minimality property of the structural confinement hull.

Let

$$X = \bigcap_{O \in \mathcal{O}_{\mathrm{str}}} \{\eta \in \mathfrak{E} \mid \mathcal{F}_O(\eta) \leq B_O\}$$

be any structurally defined subset containing $\mathfrak{R}_{\mathcal{E}}(T)$, where

$$B_O \geq C_O(T, D)$$

for every admissible structural observable.

Since

$$C_O(T, D) \leq B_O,$$

one has

$$\{\eta \mid \mathcal{F}_O(\eta) \leq C_O(T, D)\} \subseteq \{\eta \mid \mathcal{F}_O(\eta) \leq B_O\}$$

for every

$$O \in \mathcal{O}_{\mathrm{str}}.$$

Taking intersections over all admissible structural observables yields

$$\mathfrak{C}(T) \subseteq X.$$

Hence no structural confinement envelope generated by admissible structural inequalities and containing the reachable region can be strictly smaller than $\mathfrak{C}(T)$.

Therefore

$$\mathfrak{C}(T)$$

is the canonical structural confinement envelope induced by the admissible structural functionals.

□

Corollary: Insufficiency of Lieb–Robinson constraints for entanglement reachability.

Corollary 2. *For every finite evolution time*

$$T < \infty,$$

let

$$\mathfrak{E}_{\mathrm{LR}}(T) \subseteq \mathfrak{E}$$

denote the collection of entanglement structures that satisfy the Lieb–Robinson locality constraints associated with Axiom A2 alone, without imposing the additional restrictions arising from memory stability (A4) and representability compatibility (A6).

Then

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq \mathfrak{E}_{\mathrm{LR}}(T).$$

Moreover, if the additional structural restrictions induced by Axioms A4 and A6 are nontrivial, then

$$\mathfrak{R}_{\mathcal{E}}(T) \subsetneq \mathfrak{E}_{\mathrm{LR}}(T).$$

Consequently, compatibility with Lieb–Robinson locality is a necessary, but in general not sufficient, condition for dynamical entanglement reachability.

Proof. Every admissible evolution

$$\Phi_T \in \mathfrak{F}_D$$

satisfies the locality requirements of Axiom A2. Hence every reachable quantum state

$$\rho(T) \in \mathcal{R}(T)$$

obeys the corresponding Lieb–Robinson propagation constraints. Applying the entanglement-structure map gives

$$\mathcal{E}(\rho(T)) \in \mathfrak{E}_{\mathrm{LR}}(T).$$

Since

$$\mathfrak{R}_{\mathcal{E}}(T) = \mathcal{E}(\mathcal{R}(T)),$$

it follows immediately that

$$\mathfrak{R}_{\mathcal{E}}(T) \subseteq \mathfrak{E}_{\mathrm{LR}}(T).$$

Suppose now that the inclusion were an equality. Then every Lieb–Robinson-compatible entanglement structure would be dynamically reachable.

However, Theorem 1 shows that every reachable entanglement structure satisfies all structural confinement inequalities

$$\mathcal{F}_O(\eta) \leq C_O(T, D), \qquad O \in \mathcal{O}_{\mathrm{str}},$$

while Theorem 2 establishes that there exist kinematically admissible entanglement structures violating at least one such inequality.

Whenever these additional structural constraints are not implied by Axiom A2 alone, such structures remain compatible with the Lieb–Robinson locality constraints but are excluded by Axioms A4 and A6. Hence they belong to

$$\mathfrak{E}_{\mathrm{LR}}(T) \setminus \mathfrak{R}_{\mathcal{E}}(T),$$

which proves

$$\mathfrak{R}_{\mathcal{E}}(T) \subsetneq \mathfrak{E}_{\mathrm{LR}}(T).$$

Therefore Lieb–Robinson locality alone does not characterize the dynamically reachable entanglement structures.

□

Corollary: Nonredundancy of finite-memory mixing.

Corollary 3. *Let*

$$\mathfrak{R}_{\mathcal{E}}^{\mathrm{mix}}(T)$$

denote the reachable entanglement-structure set generated by admissible dynamics satisfying the exponential memory-mixing condition, and let

$$\mathfrak{R}_{\mathcal{E}}^{\mathrm{gen}}(T)$$

denote the reachable entanglement-structure set obtained by enlarging the admissible dynamical class through removal of the exponential memory-mixing requirement while leaving all remaining admissibility conditions unchanged.

Then, for every finite evolution time

$$T < \infty,$$

one has

$$\mathfrak{R}_{\mathcal{E}}^{\mathrm{mix}}(T) \subseteq \mathfrak{R}_{\mathcal{E}}^{\mathrm{gen}}(T).$$

Moreover, if the enlarged dynamical class admits entanglement-generation mechanisms supported by persistent memory correlations that are excluded by the exponential mixing condition, then

$$\mathfrak{R}_{\mathcal{E}}^{\mathrm{mix}}(T) \subsetneq \mathfrak{R}_{\mathcal{E}}^{\mathrm{gen}}(T).$$

Consequently, exponential memory mixing constitutes a structurally nonredundant restriction on dynamically reachable entanglement structures. In particular, its effect cannot, in general, be reproduced solely through the locality and representability constraints governing the remaining admissibility conditions.

Proof. Since every exponentially mixing admissible evolution is, by construction, contained in the enlarged dynamical class obtained after removing the memory-mixing requirement, one has the inclusion of admissible evolution classes. Applying the entanglement-structure map to the reachable state sets therefore yields

$$\mathfrak{R}_{\mathcal{E}}^{\mathrm{mix}}(T) \subseteq \mathfrak{R}_{\mathcal{E}}^{\mathrm{gen}}(T).$$

Suppose now that the enlarged dynamical class contains an admissible evolution whose generated entanglement structure depends essentially on persistent memory correlations that violate the exponential mixing condition. By construction, such an evolution does not belong to the exponentially mixing class. Hence its associated entanglement structure belongs to

$$\mathfrak{R}_{\mathcal{E}}^{\mathrm{gen}}(T)$$

but not to

$$\mathfrak{R}_{\mathcal{E}}^{\mathrm{mix}}(T).$$

Therefore,

$$\mathfrak{R}_{\mathcal{E}}^{\mathrm{mix}}(T) \subsetneq \mathfrak{R}_{\mathcal{E}}^{\mathrm{gen}}(T),$$

establishing the claimed strict inclusion whenever persistent memory correlations generate additional reachable entanglement structures.

Accordingly, the exponential memory-mixing condition removes genuinely reachable entanglement structures from the enlarged dynamical class and is therefore structurally nonredundant.

□

Corollary: Representability compatibility induces additional structural confinement.

Corollary 4. *Let*

$$\mathfrak{F}_D^{(\mathrm{rep})}$$

denote the class of admissible dynamics satisfying Axioms A1–A6, including the representability compatibility condition, and let

$$\mathfrak{F}_D^{(\neg\mathrm{rep})}$$

denote the enlarged dynamical class obtained by retaining all remaining admissibility conditions while omitting representability compatibility.

Define the corresponding reachable entanglement-structure sets by

$$\mathfrak{R}_{\mathcal{E}}^{(\mathrm{rep})}(T) = \left\{ \mathcal{E}(\Phi_T(\rho)) \;\middle|\; \rho \in \mathfrak{S}_D,\ \Phi_T \in \mathfrak{F}_D^{(\mathrm{rep})} \right\},$$

and

$$\mathfrak{R}_{\mathcal{E}}^{(\neg\mathrm{rep})}(T) = \left\{ \mathcal{E}(\Phi_T(\rho)) \;\middle|\; \rho \in \mathfrak{S}_D,\ \Phi_T \in \mathfrak{F}_D^{(\neg\mathrm{rep})} \right\}.$$

Then, for every finite evolution time

$$T < \infty,$$

one has

$$\mathfrak{R}_{\mathcal{E}}^{(\mathrm{rep})}(T) \subseteq \mathfrak{R}_{\mathcal{E}}^{(\neg\mathrm{rep})}(T).$$

Moreover, suppose there exists an admissible evolution

$$\Phi_T \in \mathfrak{F}_D^{(\neg\mathrm{rep})}$$

and a state
$$\rho \in \mathfrak{S}_D$$
whose induced entanglement structure fails the representability compatibility condition and therefore cannot arise from any evolution belonging to $\mathfrak{F}_D^{(\mathrm{rep})}$.

Then
$$\mathfrak{R}_{\mathcal{E}}^{(\mathrm{rep})}(T) \subsetneq \mathfrak{R}_{\mathcal{E}}^{(\neg\mathrm{rep})}(T).$$

Consequently, representability compatibility constitutes an independent structural confinement mechanism: it imposes additional restrictions on dynamically reachable entanglement structures beyond those arising from locality and finite-memory mixing alone.

Proof. Since
$$\mathfrak{F}_D^{(\mathrm{rep})} \subseteq \mathfrak{F}_D^{(\neg\mathrm{rep})},$$
every evolution satisfying representability compatibility is also admissible for the enlarged dynamical class.

Therefore,
$$\left\{\mathcal{E}(\Phi_T(\rho)) \,\middle|\, \rho \in \mathfrak{S}_D,\ \Phi_T \in \mathfrak{F}_D^{(\mathrm{rep})}\right\} \subseteq \left\{\mathcal{E}(\Phi_T(\rho)) \,\middle|\, \rho \in \mathfrak{S}_D,\ \Phi_T \in \mathfrak{F}_D^{(\neg\mathrm{rep})}\right\},$$
which is precisely
$$\mathfrak{R}_{\mathcal{E}}^{(\mathrm{rep})}(T) \subseteq \mathfrak{R}_{\mathcal{E}}^{(\neg\mathrm{rep})}(T).$$

Assume now that there exists
$$\eta^\star = \mathcal{E}(\Phi_T(\rho))$$
generated by an evolution
$$\Phi_T \in \mathfrak{F}_D^{(\neg\mathrm{rep})}$$
that is incompatible with the representability compatibility condition.

By hypothesis,
$$\eta^\star \in \mathfrak{R}_{\mathcal{E}}^{(\neg\mathrm{rep})}(T),$$
but
$$\eta^\star \notin \mathfrak{R}_{\mathcal{E}}^{(\mathrm{rep})}(T).$$

Hence
$$\mathfrak{R}_{\mathcal{E}}^{(\mathrm{rep})}(T) \neq \mathfrak{R}_{\mathcal{E}}^{(\neg\mathrm{rep})}(T),$$
and together with the previously established inclusion,
$$\mathfrak{R}_{\mathcal{E}}^{(\mathrm{rep})}(T) \subsetneq \mathfrak{R}_{\mathcal{E}}^{(\neg\mathrm{rep})}(T).$$

Therefore representability compatibility eliminates dynamically reachable entanglement structures that remain accessible in its absence, establishing that it provides an independent source of structural confinement.

□

5 Examples

The preceding sections establish a representation-independent confinement theorem for abstract entanglement-structure spaces. We now demonstrate that the theorem admits multiple mathematically distinct realizations. Each example instantiates the abstract entanglement-structure space, representability map, and structural observables differently, thereby illustrating that the confinement phenomenon is not tied to any particular model of entanglement or representation.

5.1 GHZ-depth obstruction from finite memory and representability compatibility

Example: locality-compatible GHZ structures excluded by finite memory mixing and finite representability. We first construct an explicit realization of the abstract structural confinement mechanism for a family of GHZ-type entanglement structures. The purpose of this example is not to establish a locality-only obstruction. Indeed, GHZ correlations are compatible with Lieb–Robinson propagation provided that the relevant causal region is sufficiently large. Instead, we show that when information transfer is mediated through finite-dimensional memory subsystems satisfying Axiom A4 and when the resulting states are subject to the finite representability constraint of Axiom A6, a class of GHZ structures becomes dynamically inaccessible.

Consider a finite quantum spin system

$$\mathcal{H}_N = \bigotimes_{j=1}^{N} \mathcal{H}_j,$$

with finite-dimensional local Hilbert spaces and a locality structure defined by a graph metric on the sites. The admissible evolutions satisfy the locality constraint of Axiom A2 with effective velocity

$$v_{\text{eff}}.$$

We introduce a finite-dimensional memory subsystem

$$\mathcal{H}_M$$

that mediates the generation of long-range correlations. The total Hilbert space is therefore

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_N \otimes \mathcal{H}_M.$$

The initial states are assumed to belong to the product class

$$\mathfrak{S}_{\text{prod}} = \{\rho_N \otimes \sigma_M\},$$

where

$$\rho_N = \bigotimes_{j=1}^{N} \rho_j$$

contains no initial inter-site correlations.

The target entanglement structure is the GHZ family

$$|\text{GHZ}_N\rangle = \frac{|0\rangle^{\otimes N} + |1\rangle^{\otimes N}}{\sqrt{2}},$$

with associated entanglement structure

$$\eta_{\text{GHZ}}^{(N)} = \mathcal{E}\left(|\text{GHZ}_N\rangle\langle\text{GHZ}_N|\right).$$

The relevant structural feature is the global GHZ coherence witness

$$X_{\text{GHZ}} = |0\rangle^{\otimes N}\langle 1|^{\otimes N} + |1\rangle^{\otimes N}\langle 0|^{\otimes N},$$

which satisfies

$$\|X_{\text{GHZ}}\| = 1$$

and

$$\mathrm{Tr}\left(|\mathrm{GHZ}_N\rangle\langle\mathrm{GHZ}_N|X_{\mathrm{GHZ}}\right) = 1.$$

Rather than treating X_{GHZ} itself as a structural observable, we associate to the representability constraint the induced structural functional

$$O_{\mathrm{GHZ}}(\rho) := \sup_{\|X\|\le 1} \left| \mathrm{Tr}(X\rho) - \mathrm{Tr}\left(\Pi_D^{(\mathrm{GHZ})}(X)\rho\right)\right|.$$

This quantity measures the component of operator information that lies outside the admissible representable subsystem.

For the present realization, the relevant GHZ structural space is taken to retain precisely the structural information required to distinguish representability defects of the GHZ coherence sector. More explicitly, define the GHZ-refined structural equivalence relation

$$\rho_1 \sim_{\mathrm{GHZ}} \rho_2$$

if and only if

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2)$$

and

$$O_{\mathrm{GHZ}}(\rho_1) = O_{\mathrm{GHZ}}(\rho_2).$$

The corresponding GHZ structural space is therefore the refined quotient

$$\mathfrak{E}_{\mathrm{GHZ}} := \mathfrak{S}(\mathcal{H})/\sim_{\mathrm{GHZ}}.$$

The canonical projection onto this refined structural space will be denoted by

$$\mathcal{E}_{\mathrm{GHZ}} : \mathfrak{S}(\mathcal{H}) \longrightarrow \mathfrak{E}_{\mathrm{GHZ}}.$$

The refinement

$$\mathfrak{E}_{\mathrm{GHZ}}$$

is understood as a realization-dependent structural coordinate space within the general confinement framework. It does not replace the abstract structural space

$$\mathfrak{E},$$

but specifies the structural information retained in the present GHZ realization. Accordingly, the abstract confinement construction is applied with the realization-specific structural map

$$\mathcal{E}_{\mathrm{GHZ}}$$

and the corresponding reachable region

$$\mathfrak{R}_{\mathcal{E}_{\mathrm{GHZ}}}(T) \subseteq \mathfrak{E}_{\mathrm{GHZ}}.$$

By construction, the representability-defect functional descends to this structural space, giving the induced observable

$$\mathcal{F}_{O_{\mathrm{GHZ}}} : \mathfrak{E}_{\mathrm{GHZ}} \longrightarrow \mathbb{R},$$

defined through

$$\mathcal{F}_{O_{\mathrm{GHZ}}}\left(\mathcal{E}_{\mathrm{GHZ}}(\rho)\right) = O_{\mathrm{GHZ}}(\rho).$$

Thus the GHZ representability defect is not imposed as an additional dynamical quantity, but is incorporated as a structural coordinate of the GHZ-refined entanglement structure space.

For this realization, all accessibility statements are understood with respect to the refined structural space

$$\mathfrak{E}_{\mathrm{GHZ}},$$

rather than the unrefined structural space $\mathfrak{E}$. The refinement does not modify the abstract confinement framework; it specifies the structural coordinates relevant for detecting inaccessible GHZ coherence sectors. Accordingly, the reachable structural region in this realization is denoted by

$$\mathfrak{R}_{\mathcal{E}_{\mathrm{GHZ}}}(T) \subseteq \mathfrak{E}_{\mathrm{GHZ}}.$$

In the present realization we assume that

$$O_{\mathrm{GHZ}} \in \mathcal{O}_{\mathrm{str}},$$

that is, the representability-defect functional satisfies the admissibility conditions of Definition 10. In particular, it is constant on entanglement-structure fibres, locality compatible, memory stable, representability compatible, and uniformly bounded.

Proposition: Verification of admissibility for the GHZ representability-defect functional.

Proposition 3. *For this realization, the relevant structural map is the GHZ-refined entanglement-structure map*

$$\mathcal{E}_{\mathrm{GHZ}} : \mathfrak{S}(\mathcal{H}) \longrightarrow \mathfrak{E}_{\mathrm{GHZ}},$$

where the quotient space $\mathfrak{E}_{\mathrm{GHZ}}$ is defined by retaining the representability-defect coordinate O_{GHZ} together with the original structural information. Equivalently,

$$\mathcal{E}_{\mathrm{GHZ}}(\rho_1) = \mathcal{E}_{\mathrm{GHZ}}(\rho_2)$$

implies

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2)$$

and

$$O_{\mathrm{GHZ}}(\rho_1) = O_{\mathrm{GHZ}}(\rho_2).$$

Therefore the functional

$$O_{\mathrm{GHZ}}$$

descends naturally to the refined structural space. With this refinement, the representability-defect functional

$$O_{\mathrm{GHZ}}(\rho) = \sup_{\|X\|\leq 1} |\mathrm{Tr}(X\rho) - \mathrm{Tr}(\Pi_D(X)\rho)|$$

depends only on the induced entanglement structure $\mathcal{E}(\rho)$. Assume further that the admissible dynamics satisfy Axioms A2, A4 and A6. Then

$$O_{\mathrm{GHZ}} \in \mathcal{O}_{\mathrm{str}}.$$

Proof. We verify the admissibility conditions of Definition 10.

First, by the definition of the GHZ-refined structural quotient,

$$\mathcal{E}_{\mathrm{GHZ}}(\rho_1) = \mathcal{E}_{\mathrm{GHZ}}(\rho_2)$$

implies

$$O_{\mathrm{GHZ}}(\rho_1) = O_{\mathrm{GHZ}}(\rho_2),$$

so the functional is constant on the fibres of $\mathcal{E}_{\mathrm{GHZ}}$.

Second, locality compatibility follows because Π_D acts only on the representable operator subsystem and every admissible evolution satisfies the locality propagation estimates of Lemma 2. Consequently, the operator information entering O_{GHZ} is constrained by an evolution satisfying the admissible locality hypothesis.

Third, memory stability follows from Axiom A4. Since the functional is constructed from expectation values of bounded operators, the exponential contraction of the auxiliary memory subsystem uniformly controls every memory contribution to the representability defect.

Fourth, representability compatibility is immediate from the definition of O_{GHZ}, which measures precisely the operator component discarded by the projection Π_D. Hence Axiom A6 directly provides the required stability estimate.

Finally,

$$\|X\| \leq 1$$

implies

$$|\mathrm{Tr}(X\rho) - \mathrm{Tr}(\Pi_D(X)\rho)| = |\mathrm{Tr}\left((X - \Pi_D(X))\rho\right)| \leq \|X - \Pi_D(X)\|.$$

Since Π_D is a completely positive unital map, it is contractive in the operator norm. Hence

$$\|\Pi_D(X)\| \leq \|X\| \leq 1.$$

Therefore,

$$\|X - \Pi_D(X)\| \leq \|X\| + \|\Pi_D(X)\| \leq 2.$$

Consequently,

$$0 \leq O_{\mathrm{GHZ}}(\rho) \leq 2,$$

which establishes the required uniform boundedness.

Hence every admissibility requirement of Definition 10 is satisfied.

□

The GHZ coherence witness X_{GHZ} identifies a specific contribution to this representability defect.

The generation of this coherence is assumed to proceed through the finite memory subsystem. The memory must therefore preserve a phase coherence capable of transferring the operator information associated with X_{GHZ}.

By Axiom A4, the memory channel satisfies

$$\|(\mathrm{id}_{\mathcal{H}_S} \otimes \mathcal{T}_M^s)(X) - \mathrm{Tr}_M(X) \otimes \sigma_M\| \leq Ce^{-s/\tau_M}\|X\|,$$

for every bounded operator X. Hence every component of the memory state that stores nonstationary coherence decays exponentially.

Memory-mediated realization assumption. The present realization specializes the abstract framework to the subclass of admissible evolutions for which the nonlocal component of the GHZ coherence is generated exclusively through the auxiliary memory subsystem. More precisely, we impose the following realization hypothesis.

Memory-mediated GHZ generation hypothesis. There exists a decomposition of the GHZ witness expectation value

$$\mathrm{Tr}(X_{\mathrm{GHZ}}\rho(T)) = \Delta_{\mathrm{loc}}(T) + \Delta_M(T),$$

where

- $\Delta_{\mathrm{loc}}(T)$ denotes the contribution generated entirely within the directly representable local subsystem;

- $\Delta_M(T)$ denotes the contribution whose transfer necessarily proceeds through the auxiliary memory subsystem $\mathcal{H}_M$.

The present example concerns the subclass of admissible dynamics for which all nonlocal GHZ coherence outside the directly local contribution is contained in $\Delta_M(T)$.

Under this realization hypothesis, every contribution to $\Delta_M(T)$ is carried by operators acting nontrivially on the memory degrees of freedom. Consequently, Axiom A4 applies directly to these operators.

Accordingly, the following obstruction theorem applies to the class of memory-mediated admissible evolutions satisfying Axioms A1–A6.

Applying the tensor-stable exponential mixing estimate of Axiom A4 to the memory-supported operators defining $\Delta_M(T)$ therefore yields

$$\Delta_M(T) \leq C_M e^{-T/\tau_M},$$

where the constant C_M depends only on the finite-dimensional memory realization and the initial memory state.

Thus, within the present realization, every nonlocal GHZ coherence contribution transmitted through the auxiliary memory subsystem decays exponentially.

We now impose a concrete representability realization of Axioms A5–A6 appropriate to the present GHZ example. The abstract framework does not require a particular form for the representable subsystem $\mathcal{A}_D$; rather, Axiom A5 only assumes the existence of an admissible operator subsystem.

For this realization, we choose the representative subsystem in which only the degrees of freedom contained in

$$\Lambda_D \subseteq \{1, \ldots, N\}$$

remain accessible at representability scale D. The corresponding representable operator subsystem is defined by

$$\mathcal{A}_D^{(\mathrm{GHZ})} = \mathcal{B}\left(\bigotimes_{j\in\Lambda_D} \mathcal{H}_j\right) \otimes \mathbf{1}_{\Lambda_D^c}.$$

For notational simplicity, we write

$$\mathcal{A}_D^{(\mathrm{GHZ})} \equiv \mathcal{A}_D$$

within this realization only.

The associated completely positive unital idempotent projection for this choice of representability subsystem is

$$\Pi_D^{(\mathrm{GHZ})} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{(\mathrm{GHZ})},$$

with

$$\left(\Pi_D^{(\mathrm{GHZ})}\right)^2 = \Pi_D^{(\mathrm{GHZ})}.$$

Again, this projection is a concrete realization of the abstract representability map of Axiom A6 rather than a universal choice of representability structure. For every finite representability scale satisfying

$$\Lambda_D \subsetneq \{1, \ldots, N\},$$

the GHZ coherence operator is not contained in the representable subsystem:

$$X_{\mathrm{GHZ}} \notin \mathcal{A}_D.$$

Indeed, every element of the representable algebra

$$\mathcal{A}_D = \mathcal{B}\left(\bigotimes_{j\in\Lambda_D} \mathcal{H}_j\right) \otimes \mathbf{1}_{\Lambda_D^c}$$

acts trivially on the complementary degrees of freedom Λ_D^c. Therefore any operator belonging to $\mathcal{A}_D$ has matrix elements between basis states that differ only through the degrees of freedom contained in Λ_D.

By contrast,

$$X_{\mathrm{GHZ}} = |0\rangle^{\otimes N}\langle 1|^{\otimes N} + |1\rangle^{\otimes N}\langle 0|^{\otimes N}$$

connects two configurations whose support differs on every site. Hence it requires nontrivial operator support on the entire tensor factor

$$\bigotimes_{j=1}^{N} \mathcal{H}_j.$$

Therefore, whenever

$$\Lambda_D \subsetneq \{1, \dots, N\},$$

one necessarily has

$$X_{\mathrm{GHZ}} \notin \mathcal{A}_D.$$

The exclusion of X_{GHZ} from the representable operator subsystem must be converted into a quantitative defect estimate. For the present concrete realization, this follows directly from the structure of the representability projection.

The projection

$$\Pi_D^{(\mathrm{GHZ})} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{(\mathrm{GHZ})}$$

is chosen as the conditional expectation onto the accessible subsystem, namely

$$\Pi_D^{(\mathrm{GHZ})}(X) = \mathrm{Tr}_{\Lambda_D^c}(X) \otimes \frac{\mathbf{1}_{\Lambda_D^c}}{\dim(\mathcal{H}_{\Lambda_D^c})}.$$

For the GHZ coherence operator,

$$X_{\mathrm{GHZ}} = |0\rangle^{\otimes N}\langle 1|^{\otimes N} + |1\rangle^{\otimes N}\langle 0|^{\otimes N},$$

the partial trace over any omitted degree of freedom vanishes because

$$\mathrm{Tr}\left(|0\rangle\langle 1|\right) = 0$$

and

$$\mathrm{Tr}\left(|1\rangle\langle 0|\right) = 0.$$

Therefore, whenever

$$\Lambda_D \subsetneq \{1, \dots, N\},$$

one has

$$\mathrm{Tr}_{\Lambda_D^c}(X_{\mathrm{GHZ}}) = 0,$$

and consequently

$$\Pi_D^{(\mathrm{GHZ})}(X_{\mathrm{GHZ}}) = 0.$$

Hence the GHZ witness sector is orthogonal to the retained representable operator subsystem in this realization, and

$$\left|\mathrm{Tr}\left(X_{\mathrm{GHZ}}\rho\right) - \mathrm{Tr}\left(\Pi_D^{(\mathrm{GHZ})}(X_{\mathrm{GHZ}})\rho\right)\right| = \left|\mathrm{Tr}\left(X_{\mathrm{GHZ}}\rho\right)\right|.$$

Under this condition,

$$\left|\mathrm{Tr}\left(X_{\mathrm{GHZ}}\rho\right) - \mathrm{Tr}\left(\Pi_D(X_{\mathrm{GHZ}})\rho\right)\right| = \left|\mathrm{Tr}\left(X_{\mathrm{GHZ}}\rho\right)\right|.$$

Since

$$\|X_{\mathrm{GHZ}}\| = 1,$$

the witness contribution is one of the terms included in the supremum defining O_{GHZ}, and therefore

$$|\mathrm{Tr}\,(X_{\mathrm{GHZ}}\rho)| \leq O_{\mathrm{GHZ}}(\rho).$$

Hence the GHZ coherence detected by X_{GHZ} is directly controlled by the representability-defect functional.

Therefore the representability compatibility condition of Axiom A6 gives

$$\left\|\Pi_D^{(\mathrm{GHZ})} \circ \Phi_T \circ \Pi_D^{(\mathrm{GHZ})}(\rho) - \Phi_T \circ \Pi_D^{(\mathrm{GHZ})}(\rho)\right\| \leq \varepsilon_D(T).$$

By the duality between the trace norm and the operator norm, the representability estimate of Axiom A6 implies that for every operator X satisfying $\|X\| \leq 1$,

$$|\mathrm{Tr}\,[X\,(\Pi_D \circ \Phi_T \circ \Pi_D(\rho) - \Phi_T \circ \Pi_D(\rho))]| \leq \varepsilon_D(T).$$

The Axiom A6 compatibility estimate controls the deviation between the evolution of the representable component and the projection of the evolved representable component. Accordingly, define the representability-defect state

$$\delta_D(T,\rho) := \Phi_T(\Pi_D(\rho)) - \Pi_D(\Phi_T(\Pi_D(\rho)))\,.$$

By Axiom A6,

$$\|\delta_D(T,\rho)\| \leq \varepsilon_D(T).$$

For every operator X satisfying $\|X\| \leq 1$, the dual norm estimate gives

$$|\mathrm{Tr}\,(X\delta_D(T,\rho))| \leq \varepsilon_D(T).$$

The representability defect relevant to the present realization is therefore the component of the evolved state carried by $\delta_D(T,\rho)$. Hence the induced GHZ representability functional is evaluated on the representability-defect sector,

$$O_{\mathrm{GHZ}}(\rho(T)) \equiv \sup_{\|X\|\leq 1} |\mathrm{Tr}\,(X\delta_D(T,\rho))|\,.$$

Consequently,

$$O_{\mathrm{GHZ}}(\rho(T)) \leq \varepsilon_D(T).$$

This identification makes explicit that the structural defect controlled by Axiom A6 is the representability-loss component of the evolved state rather than the full state $\rho(T)$. Equivalently, the component of the evolved state that is not captured by the representable subsystem is bounded by the representability error.

In particular, since the GHZ coherence witness satisfies

$$\|X_{\mathrm{GHZ}}\| = 1,$$

its contribution to the representability defect obeys

$$\Delta_D(T) \leq \varepsilon_D(T),$$

where $\Delta_D(T)$ denotes the GHZ coherence contribution that cannot be reconstructed from information retained inside the representable subsystem.

The GHZ coherence contribution is obtained by restricting the representability-defect functional to the witness sector generated by X_{GHZ}. Since

$$\|X_{\mathrm{GHZ}}\| = 1,$$

this sector is contained in the supremum defining O_{GHZ}, and therefore the corresponding structural contribution is bounded by the same estimate.

Combining the finite-memory mixing estimate with representability compatibility yields the total GHZ coherence bound.

Proposition 4 (GHZ obstruction from finite memory and representability)**.** *For every memory-mediated admissible evolution satisfying Axioms A1–A6,*

$$\mathcal{F}_{O_{\mathrm{GHZ}}}\left(\mathcal{E}(\rho(T))\right) \leq C_M e^{-T/\tau_M} + \varepsilon_D(T).$$

Consequently, any GHZ structure satisfying

$$\left|\mathrm{Tr}\left(X_{\mathrm{GHZ}}\rho_{\mathrm{target}}\right)\right| > C_M e^{-T/\tau_M} + \varepsilon_D(T)$$

cannot belong to the GHZ-refined reachable structural set

$$\mathfrak{R}_{\mathcal{E}_{\mathrm{GHZ}}}(T) \subseteq \mathfrak{C}_{\mathrm{GHZ}}.$$

Proof. The first contribution follows from Axiom A4. Any coherence stored in the finite memory subsystem decays exponentially toward the stationary memory state. Hence the maximum contribution that can be transferred through the memory channel is bounded by

$$C_M e^{-T/\tau_M}.$$

The second contribution follows from Axiom A6 together with the dual-norm estimate above. The representability defect state controlled by Axiom A6 induces the structural functional

$$\mathcal{F}_{O_{\mathrm{GHZ}}}$$

through the witness-sector contribution described above. Therefore this structural defect is bounded by the representability error

$$\varepsilon_D(T).$$

Using the GHZ witness incompatibility condition established above, the GHZ coherence contribution satisfies

$$\left|\mathrm{Tr}\left(X_{\mathrm{GHZ}}\rho(T)\right)\right| \leq O_{\mathrm{GHZ}}(\rho(T)).$$

The representability estimate and the memory-mixing estimate then imply

$$O_{\mathrm{GHZ}}(\rho(T)) \leq C_M e^{-T/\tau_M} + \varepsilon_D(T).$$

Combining these two inequalities gives

$$\left|\mathrm{Tr}\left(X_{\mathrm{GHZ}}\rho(T)\right)\right| \leq C_M e^{-T/\tau_M} + \varepsilon_D(T).$$

Therefore every target structure requiring a larger GHZ coherence lies outside the reachable structural region.

□

This obstruction is compatible with Lieb–Robinson locality. In particular, the GHZ structure is not excluded because information propagates faster than the allowed velocity v_{eff}. Rather, the causal region permitted by Axiom A2 contains the target structure, but the required global coherence cannot be maintained by the finite memory dynamics and cannot be retained within the finite representability subsystem.

Hence the GHZ family provides an explicit realization of the general structural confinement theorem within the GHZ-refined structural space:

$$\eta^{(N)}_{\mathrm{GHZ}} \in \mathfrak{C}_{\mathrm{GHZ}} \setminus \mathfrak{R}_{\mathcal{E}_{\mathrm{GHZ}}}(T)$$

whenever

$$1 > C_M e^{-T/\tau_M} + \varepsilon_D(T).$$

The obstruction therefore arises from the combined action of locality, finite-memory mixing, and representability compatibility, demonstrating that kinematic admissibility is strictly weaker than dynamical realizability.

This example realizes the abstract confinement hull of Theorem 3 within the refined GHZ structural space: the GHZ coherence functional defines a structural inequality whose violation identifies GHZ entanglement structures outside the dynamically reachable region

$$\mathfrak{R}_{\mathcal{E}_{\mathrm{GHZ}}}(T).$$

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5.2 Tensor-network realization of finite representability obstruction

Example: tensor-network representability as a realization of structural confinement. We next construct a second realization of the abstract confinement mechanism developed in Sections 1–4. Unlike the preceding GHZ example, the present realization does not concern multipartite coherence or memory-mediated generation of long-range correlations. Instead, the obstruction arises from the representability structure itself.

The purpose of this example is to show that Axiom A6 excludes a class of entanglement structures whose faithful description necessarily requires operator information lying outside a finite MPO-adapted representability subsystem. Thus the confinement mechanism is now controlled by finite representability rather than by finite-memory coherence.

Throughout this example we adopt a matrix-product-operator (MPO) realization of the representable operator subsystems introduced abstractly in Axiom A5. No statement is made concerning the optimality or uniqueness of this realization; it merely provides one explicit instantiation of the abstract framework.

Physical realization. Consider a finite quantum lattice

$$\mathcal{H}_N = \bigotimes_{j=1}^{N} \mathcal{H}_j,$$

where every local Hilbert space is finite dimensional.

The admissible dynamics satisfy Axioms A1–A6.

Unlike Example 1, no assumption is made concerning GHZ-type coherence or memory-mediated generation of correlations. The only structural restriction considered here is finite operator representability.

We assume that the admissible initial states belong to the class

$$\mathfrak{S}_{\mathrm{TN}} \subseteq \mathfrak{S}(\mathcal{H}_N),$$

consisting of states whose physically admissible descriptions are compatible with the finite representability scale prescribed by Axiom A5.

The precise internal parametrization of this class is intentionally left unspecified, since the abstract theory requires only the existence of the representable operator subsystem rather than any particular tensor decomposition.

Tensor-network realization of the entanglement structure. The corresponding entanglement-structure space is denoted

$$\mathfrak{E}_{\mathrm{TN}},$$

whose elements classify entanglement structures according to their representation within the admissible tensor-network description.

The tensor-network structural space is not identified with the set of bond dimensions or with a tensor-network complexity measure alone. Rather, $\mathfrak{E}_{\mathrm{TN}}$ is defined through the information retained by the dual representable observable subsystem associated with $\mathcal{A}_D^{\mathrm{MPO}}$. Let

$$(\Pi_D^{\mathrm{MPO}})^* : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{B}(\mathcal{H}_N)$$

denote the dual map acting on states under the chosen operator-space pairing. The physically retained observables are identified with the range of the Heisenberg-picture representability projection, namely

$$\mathrm{Ran}\left(\Pi_D^{\mathrm{MPO}}\right) = \mathcal{A}_D^{\mathrm{MPO}}.$$

Equivalently, the retained observables satisfy

$$A = \Pi_D^{\mathrm{MPO}}(A).$$

The tensor-network structural equivalence relation is defined by

$$\rho_1 \sim_{\mathrm{TN}} \rho_2$$

if and only if the complete representability-defect data associated with the MPO-adapted projection coincide, namely

$$(\Pi_D^{\mathrm{MPO}})^*(\rho_1) = (\Pi_D^{\mathrm{MPO}})^*(\rho_2),$$

and

$$\left\|\rho_1 - (\Pi_D^{\mathrm{MPO}})^*(\rho_1)\right\|_1 = \left\|\rho_2 - (\Pi_D^{\mathrm{MPO}})^*(\rho_2)\right\|_1.$$

Equivalently, the structural quotient retains precisely the information required to evaluate the dual representability defect observable. The resulting tensor-network structural space is therefore

$$\mathfrak{E}_{\mathrm{TN}} = \mathfrak{S}(\mathcal{H}_N)/\sim_{\mathrm{TN}}.$$

By construction, every structural functional depending only on the retained dual observable algebra is constant on the fibres of the entanglement-structure map

$$\mathcal{E} : \mathfrak{S}(\mathcal{H}_N) \longrightarrow \mathfrak{E}_{\mathrm{TN}},$$

and therefore induces a well-defined functional on $\mathfrak{E}_{\mathrm{TN}}$.

As in Section 4, the entanglement-structure map is written

$$\mathcal{E} : \mathfrak{S}(\mathcal{H}_N) \longrightarrow \mathfrak{E}_{\mathrm{TN}}.$$

In the present example, $\mathfrak{E}_{\mathrm{TN}}$ is introduced as one admissible realization of the abstract entanglement-structure space of Sections 1–4. Specifically, the underlying structural quotient is chosen so that the information contained in the MPO-adapted representable observable subsystem of $(\Pi_D^{\mathrm{MPO}})^*$ constitutes the structural invariant data. Consequently, any structural functional constructed solely from these retained observables is constant on the fibres of $\mathcal{E}$ by construction. This choice is made solely for the present realization and is not intended to define a unique physical entanglement-structure space.

MPO-adapted representable operator subsystem. For every representability scale D, let

$$\mathcal{A}_D^{\mathrm{MPO}} \subseteq \mathcal{B}(\mathcal{H}_N)$$

denote a finite-scale operator subsystem generated by a prescribed family of admissible MPO tensors. The resulting space is assumed to satisfy the operator-system requirements of Axiom A5 and the compatibility requirements of Axiom A6. Importantly, $\mathcal{A}_D^{\mathrm{MPO}}$ is not identified with the nonlinear set of all MPOs of bond dimension bounded by D. Rather, the MPO tensors provide a concrete generating description of an operator subsystem for which the completely positive unital idempotent representability projection required by Axiom A6 exists. In the present realization, this subsystem is interpreted as an MPO-adapted representable operator subsystem. Thus the matrix-product- operator language serves only as a concrete realization of the abstract representability structure rather than as its definition.

The associated representability map is the completely positive unital idempotent projection

$$\Pi_D^{\mathrm{MPO}} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{\mathrm{MPO}},$$

satisfying

$$(\Pi_D^{\mathrm{MPO}})^2 = \Pi_D^{\mathrm{MPO}},$$

where the projection is precisely the completely positive unital idempotent map postulated by Axiom A6 for the chosen admissible operator subsystem.

No particular bond-dimension parametrization is assumed. The parameter D denotes the abstract representability scale of Axioms A5–A6. The present matrix-product-operator realization merely provides one admissible realization of the corresponding operator subsystem and does not identify $\mathcal{A}_D^{\mathrm{MPO}}$ with the collection of all finite-bond-dimension MPOs.

Accordingly, the obstruction considered here is not the existence of a finite MPO representation per se, but the restriction to the fixed admissible representability subsystem associated with the abstract scale D.

Dual representability defect structural observable. The abstract framework of Sections 1–4 does not prescribe a unique structural observable for a given representability subsystem. In the present MPO realization, the structural observable is constructed directly from the dual action of the representability projection.

The operator subsystem

$$\mathcal{A}_D^{\mathrm{MPO}} \subseteq \mathcal{B}(\mathcal{H}_N)$$

is equipped with the completely positive unital idempotent projection

$$\Pi_D^{\mathrm{MPO}} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{\mathrm{MPO}}.$$

The corresponding Schrödinger-picture dual map is defined by

$$(\Pi_D^{\mathrm{MPO}})^* : \mathfrak{S}(\mathcal{H}_N) \longrightarrow \mathfrak{S}(\mathcal{H}_N),$$

through the duality relation

$$\mathrm{Tr}\left[(\Pi_D^{\mathrm{MPO}})^*(\rho) A\right] = \mathrm{Tr}\left[\rho\, \Pi_D^{\mathrm{MPO}}(A)\right],$$

for every state $\rho \in \mathfrak{S}(\mathcal{H}_N)$ and every operator $A \in \mathcal{B}(\mathcal{H}_N)$. The representability defect of a state is therefore quantified by the component that remains distinguishable from its representable image through the admissible dual operator sector. We define the MPO-adapted structural observable by

$$O_{\mathrm{TN}}^{(D)}(\rho) = \sup_{\substack{A \in \mathcal{B}(\mathcal{H}_N) \\ \|A\| \leq 1}} \left| \mathrm{Tr}\left[\left(\rho - (\Pi_D^{\mathrm{MPO}})^*(\rho)\right) A\right] \right|.$$

Thus $O_{\mathrm{TN}}^{(D)}$ measures the norm distance between a state and its representable image, equivalently quantifying the component of the state that is discarded by the admissible MPO representability projection. The supremum is taken over the full dual operator space because the defect is a state-space distance rather than an expectation-value difference restricted to the retained observable sector.

The structural observable is therefore determined intrinsically by the chosen representability subsystem rather than introduced as an independent functional. The resulting defect measures the component of the state removed by the representability projection, quantified by the full dual state-space norm.

Finite operator-subsystem defect structural observable. The structural observable associated with the present realization is defined directly from the dual representability projection rather than introduced through an additional compatibility assumption.

Let
$$(\Pi_D^{\mathrm{MPO}})^* : \mathfrak{S}(\mathcal{H}_N) \longrightarrow \mathfrak{S}(\mathcal{H}_N)$$
denote the dual map characterized by
$$\mathrm{Tr}\big((\Pi_D^{\mathrm{MPO}})^*(\rho)\, A\big) = \mathrm{Tr}\big(\rho\, \Pi_D^{\mathrm{MPO}}(A)\big)$$
for every state
$$\rho \in \mathfrak{S}(\mathcal{H}_N)$$
and every observable
$$A \in \mathcal{B}(\mathcal{H}_N).$$
The tensor-network structural observable is defined by
$$O_{\mathrm{TN}} : \mathfrak{S}(\mathcal{H}_N) \longrightarrow \mathbb{R},$$
with
$$O_{\mathrm{TN}}(\rho) = \sup_{\substack{A\in\mathcal{B}(\mathcal{H}_N)\\ \|A\|\le 1}} \left|\mathrm{Tr}\left[\big(\rho - (\Pi_D^{\mathrm{MPO}})^*(\rho)\big)A\right]\right|.$$
Equivalently, by duality between the operator norm and trace norm,
$$O_{\mathrm{TN}}(\rho) = \left\|\rho - (\Pi_D^{\mathrm{MPO}})^*(\rho)\right\|_1.$$
Thus $O_{\mathrm{TN}}(\rho)$ measures the maximal distinguishability between the physical state and its projected representable description. The defect quantifies the portion of the state that cannot be retained inside the admissible representability subsystem.

Since the structural quotient defining
$$\mathfrak{E}_{\mathrm{TN}}$$
is generated by the MPO-adapted representable observable subsystem, O_{TN} depends only on the induced entanglement structure. This follows because the dual projection $(\Pi_D^{\mathrm{MPO}})^*$ is determined entirely by the representability subsystem $\mathcal{A}_D^{\mathrm{MPO}}$; therefore states belonging to the same quotient class have identical projected representable components. Accordingly, the induced structural functional
$$\mathcal{F}_{O_{\mathrm{TN}}} : \mathfrak{E}_{\mathrm{TN}} \longrightarrow \mathbb{R}$$
is defined by
$$\mathcal{F}_{O_{\mathrm{TN}}}\big(\mathcal{E}(\rho)\big) = O_{\mathrm{TN}}(\rho).$$

Lemma: Dual representability defect bound.

Lemma 4. *For every admissible evolution*

$$\Phi_T \in \mathfrak{F}_D$$

and every state

$$\rho \in \mathfrak{S}(\mathcal{H}_N),$$

the dual representability defect satisfies

$$O_{\mathrm{TN}}(\rho(T)) \leq \sup_{\substack{A\in\mathcal{B}(\mathcal{H}_N)\\ \|A\|\leq 1}} \left|\mathrm{Tr}\left[\left(\Pi_D^{\mathrm{MPO}} \circ \Phi_T \circ \Pi_D^{\mathrm{MPO}} - \Phi_T \circ \Pi_D^{\mathrm{MPO}}\right)(\rho)\, A\right]\right|.$$

Consequently,

$$O_{\mathrm{TN}}(\rho(T))$$

is controlled by the representability compatibility defect appearing in Axiom A6.

Proof. By definition,

$$O_{\mathrm{TN}}(\rho(T)) = \sup_{\substack{A\in\mathcal{B}(\mathcal{H}_N)\\ \|A\|\leq 1}} \left|\mathrm{Tr}\left[\left(\rho(T) - (\Pi_D^{\mathrm{MPO}})^*(\rho(T))\right) A\right]\right|.$$

The duality relation defining $(\Pi_D^{\mathrm{MPO}})^*$ expresses the representability defect as the difference between the physical state and its projected representable image. Using the duality relation, the defect functional can be bounded by the dual action of the compatibility defect appearing in Axiom A6.

$$\rho(T) - (\Pi_D^{\mathrm{MPO}})^*(\rho(T)) \preceq \left(\Pi_D^{\mathrm{MPO}} \circ \Phi_T \circ \Pi_D^{\mathrm{MPO}} - \Phi_T \circ \Pi_D^{\mathrm{MPO}}\right)^* (\rho).$$

The corresponding state-space norm is evaluated through the full dual operator space.

For every bounded observable $A \in \mathcal{B}(\mathcal{H}_N)$, the corresponding expectation-value difference is bounded by the representability compatibility defect appearing in Axiom A6.

By the duality relation between the chosen state-space norm and the operator norm appearing in the supremum, the expectation-value difference is bounded by the norm defect appearing in Axiom A6. Taking the supremum over all admissible observables yields the stated bound. □

Proposition: Admissibility of the dual representability defect functional. The following proposition establishes that the MPO-adapted structural observable constructed from the MPO-adapted representable observable subsystem satisfies the requirements of $\mathcal{O}_{\mathrm{str}}$.

Proposition 5. *Let*

$$O_{\mathrm{TN}} : \mathfrak{S}(\mathcal{H}_N) \longrightarrow \mathbb{R}$$

be the dual representability defect functional defined by

$$O_{\mathrm{TN}}(\rho) = \sup_{\substack{A\in\mathcal{B}(\mathcal{H}_N)\\ \|A\|\leq 1}} \left|\mathrm{Tr}\left[\left(\rho - (\Pi_D^{\mathrm{MPO}})^*(\rho)\right) A\right]\right|.$$

Equivalently,

$$O_{\mathrm{TN}}(\rho) = \left\|\rho - (\Pi_D^{\mathrm{MPO}})^*(\rho)\right\|_1.$$

Assume that the admissible dynamics and representability structure satisfy Axioms A2, A4, A5 and A6. Then

$$O_{\mathrm{TN}} \in \mathcal{O}_{\mathrm{str}}.$$

Proof. We verify each admissibility condition of Definition 10.

First, by construction of the tensor-network structural quotient, two states satisfying

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2)$$

have identical representability-defect data associated with

$$(\Pi_D^{\mathrm{MPO}})^*.$$

Therefore,

$$O_{\mathrm{TN}}(\rho_1) = O_{\mathrm{TN}}(\rho_2).$$

Hence O_{TN} is constant on the fibres of $\mathcal{E}$ and induces a well-defined structural functional

$$\mathcal{F}_{O_{\mathrm{TN}}} : \mathfrak{E}_{\mathrm{TN}} \longrightarrow \mathbb{R}.$$

Second, locality compatibility follows from the fact that the structural functional is constructed from the admissible representability subsystem associated with $\mathcal{A}_D^{\mathrm{MPO}}$ and from dynamics satisfying the locality assumptions of Axiom A2. No additional locality property of the projection itself is required.

Third, memory stability follows because O_{TN} depends only on the physical state and the fixed representability structure. The auxiliary memory degrees of freedom introduced in Axiom A3 do not appear as independent arguments of the functional.

Any influence of memory degrees of freedom on the physical evolution is therefore controlled by the admissible dynamics and the mixing condition of Axiom A4.

Fourth, representability compatibility follows from the dual defect estimate proved in Lemma 4, which gives

$$O_{\mathrm{TN}}(\rho(T)) \leq \varepsilon_D(T)$$

for every admissible evolution.

Finally, uniform boundedness follows from the boundedness of the admissible observable family together with finite dimensionality of $\mathcal{H}_N$. Since

$$\|A\| \leq 1$$

in the defining supremum, Hölder duality gives

$$O_{\mathrm{TN}}(\rho) = \left\|\rho - (\Pi_D^{\mathrm{MPO}})^*(\rho)\right\|_1,$$

which is finite because $(\Pi_D^{\mathrm{MPO}})^*$ is a bounded linear map on the finite-dimensional state space.

Therefore all admissibility requirements of Definition 10 are satisfied, and hence

$$O_{\mathrm{TN}} \in \mathcal{O}_{\mathrm{str}}.$$

□

Realization of the representability compatibility axiom. The present realization interprets the abstract representable operator subsystem supplied by Axiom A5 through an MPO-adapted admissible operator subsystem

$$\mathcal{A}_D^{\mathrm{MPO}} \subseteq \mathcal{B}(\mathcal{H}_N),$$

together with the associated completely positive unital idempotent projection

$$\Pi_D^{\mathrm{MPO}} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{\mathrm{MPO}}.$$

Accordingly, the compatibility estimate of Axiom A6 becomes

$$\left\|\Pi_D^{\mathrm{MPO}} \circ \Phi_T \circ \Pi_D^{\mathrm{MPO}}(\rho) - \Phi_T \circ \Pi_D^{\mathrm{MPO}}(\rho)\right\| \leq \varepsilon_D(T),$$

for every admissible evolution

$$\Phi_T \in \mathfrak{F}_D$$

and every state

$$\rho \in \mathfrak{S}(\mathcal{H}_N).$$

Thus the chosen MPO-adapted representable subsystem inherits the compatibility property required by Axiom A6, preserving the information contained within the representable sector up to the approximation error $\varepsilon_D(T)$.

We now translate this compatibility estimate into a structural bound for the induced tensor-network functional. The dual representability defect estimate established in Lemma 4 directly gives the corresponding structural bound

$$O_{\mathrm{TN}}(\rho(T)) \leq \varepsilon_D(T).$$

Thus the confinement estimate follows from the construction of the dual observable together with the compatibility property of Axiom A6, without introducing any additional realization-specific compatibility assumption.

Therefore,

$$O_{\mathrm{TN}}(\rho(T)) \leq \varepsilon_D(T),$$

so the dual representability defect remains uniformly bounded by the finite-scale compatibility error prescribed by Axiom A6. Hence the representability defect of every reachable state is bounded by the finite-scale compatibility error.

Equivalently, the component of the evolved state that cannot be encoded within the chosen MPO realization of the admissible representable operator subsystem is uniformly bounded by the representability error.

This estimate is an MPO realization of the abstract finite representability compatibility axiom. Unlike the preceding GHZ example, the confinement mechanism does not depend on the behavior of any distinguished coherence witness or multipartite entanglement measure. Rather, the obstruction arises because operator information lying outside the finite representable subsystem is not retained within the admissible tensor-network description.

Accordingly, the structural functional

$$\mathcal{F}_{O_{\mathrm{TN}}}$$

satisfies

$$\mathcal{F}_{O_{\mathrm{TN}}}\left(\mathcal{E}(\rho(T))\right) = O_{\mathrm{TN}}\big(\rho(T)\big) \leq \varepsilon_D(T).$$

Thus every admissible evolution obeying Axioms A1–A6 remains confined to the subset of entanglement structures whose tensor-network representability defect is bounded by the approximation error $\varepsilon_D(T)$.

The resulting confinement therefore reflects a limitation of finite representability rather than a limitation of entanglement generation itself. States may remain compatible with locality and finite-time dynamics while nevertheless possessing structural features that cannot be faithfully represented within the admissible finite-scale MPO-adapted operator subsystem.

Proposition: Tensor-network representability obstruction.

Proposition 6. *For every admissible evolution satisfying Axioms A1–A6,*

$$\mathcal{F}_{O_{\mathrm{TN}}}\left(\mathcal{E}(\rho(T))\right) \leq \varepsilon_D(T).$$

Consequently, every entanglement structure

$$\eta \in \mathfrak{E}_{\mathrm{TN}}$$

satisfying

$$\mathcal{F}_{O_{\mathrm{TN}}}(\eta) > \varepsilon_D(T)$$

cannot belong to the reachable structural set

$$\mathfrak{R}_{\mathcal{E}}(T).$$

Proof. By Lemma 4, the representability compatibility estimate of Axiom A6 implies

$$O_{\mathrm{TN}}(\rho(T)) \leq \varepsilon_D(T).$$

By construction of the induced structural functional,

$$\mathcal{F}_{O_{\mathrm{TN}}}\left(\mathcal{E}(\rho(T))\right) = O_{\mathrm{TN}}(\rho(T)).$$

Hence

$$\mathcal{F}_{O_{\mathrm{TN}}}\left(\mathcal{E}(\rho(T))\right) \leq \varepsilon_D(T).$$

Now let

$$\eta \in \mathfrak{E}_{\mathrm{TN}}$$

satisfy

$$\mathcal{F}_{O_{\mathrm{TN}}}(\eta) > \varepsilon_D(T).$$

If

$$\eta \in \mathfrak{R}_{\mathcal{E}}(T),$$

then there would exist an admissible evolution producing a state $\rho(T)$ with

$$\mathcal{E}(\rho(T)) = \eta.$$

However, the preceding estimate yields

$$\mathcal{F}_{O_{\mathrm{TN}}}(\eta) = \mathcal{F}_{O_{\mathrm{TN}}}\left(\mathcal{E}(\rho(T))\right) \leq \varepsilon_D(T),$$

contradicting the assumed inequality.
Therefore

$$\eta \notin \mathfrak{R}_{\mathcal{E}}(T).$$

□

Explicit forbidden structural family. The preceding proposition identifies a concrete family of forbidden entanglement structures. Let

$$\{\eta_\lambda\}_{\lambda\in\Lambda} \subset \mathfrak{E}_{\mathrm{TN}}$$

be any family satisfying

$$\mathcal{F}_{O_{\mathrm{TN}}}(\eta_\lambda) > \varepsilon_D(T)$$

for every $\lambda \in \Lambda$. Then, by Proposition 6,

$$\eta_\lambda \notin \mathfrak{R}_{\mathcal{E}}(T)$$

for every $\lambda \in \Lambda$. Thus the finite representability obstruction excludes an entire class of tensor-network structures whose structural information cannot be retained inside the admissible operator subsystem. The family is defined by violation of the representability threshold, rather than by a particular choice of tensor-network complexity parameter.

A canonical realization of such a family is obtained by choosing states whose reduced operator data are asymptotically orthogonal to the admissible representable subsystem. For example, let $\{\rho_\lambda\}$ be a family satisfying

$$\lim_{\lambda\to\infty} \left\|\rho_\lambda - (\Pi_D^{\mathrm{MPO}})^*(\rho_\lambda)\right\|_1 = \delta_D > 0.$$

Then the corresponding structural classes

$$\eta_\lambda = \mathcal{E}(\rho_\lambda)$$

satisfy

$$\mathcal{F}_{O_{\mathrm{TN}}}(\eta_\lambda) \geq \delta_D,$$

and therefore become forbidden whenever

$$\delta_D > \varepsilon_D(T).$$

Thus the obstruction excludes not a particular tensor-network ansatz, but any structural family whose operator information remains separated from the admissible representability subsystem.

The obstruction established above is fundamentally different from the GHZ realization of Example 1. In particular, no restriction is imposed by insufficient causal propagation or by decay of multipartite coherence. The admissible dynamics may remain fully compatible with the locality assumptions of Axiom A2 and the memory dynamics of Axiom A4.

Instead, the confinement arises because finite representability excludes operator information lying outside the chosen MPO realization of the admissible representable operator subsystem. Consequently, there exist entanglement structures whose complete structural description requires operator information that cannot be faithfully retained within the representable subsystem, even though the underlying dynamics remain kinematically admissible.

The obstruction therefore reflects a limitation of finite structural representation rather than a limitation of entanglement generation itself. In particular, a target state may satisfy all locality constraints while nevertheless lying outside the dynamically reachable structural region because its structural information cannot be encoded within the admissible representability scale.

Hence the present realization provides a second explicit instantiation of the abstract structural confinement theorem. Indeed,

$$\eta \notin \mathfrak{R}_{\mathcal{E}}(T)$$

whenever

$$\mathcal{F}_{O_{\mathrm{TN}}}(\eta) > \varepsilon_D(T).$$

Thus the finite operator-subsystem defect defines a structural inequality whose violation identifies entanglement structures lying outside the dynamically reachable region.

This example therefore realizes the abstract confinement hull of Theorem 3. Unlike the preceding GHZ realization, the present obstruction is generated entirely by the finite representability structure embodied in Axioms A5–A6.

Finally, the choice of matrix-product operators serves only as one concrete realization of the abstract representable operator subsystem postulated by Axioms A5 and A6. The confinement theorem itself is independent of this particular realization and does not rely on any specific characterization of finite-bond-dimension tensor-network classes. Any admissible operator subsystem equipped with the completely positive unital idempotent projection required by Axiom A6 yields an analogous representability obstruction. The confinement mechanism is therefore a structural consequence of the abstract axiomatic framework rather than of any particular tensor-network formalism.

5.3 Quantum error-correcting code and logical-structure obstruction.

Example: protected logical structures excluded by finite logical representability We next construct an independent realization of the abstract structural confinement mechanism developed in Sections 1–4 within the setting of quantum error correction. Unlike the preceding GHZ realization, the present example does not concern multipartite coherence or memory-mediated generation of long-range correlations. Likewise, unlike the tensor-network realization, the obstruction is not formulated in terms of finite tensor-network descriptions.

Instead, the relevant structural objects are protected logical information and the corresponding logical operator structures associated with encoded quantum states. The purpose of this example is to show that, for logical-code realizations satisfying an additional logical representability growth condition, the observable-level form of Axiom A6 excludes a class of logical entanglement structures whose faithful representation requires logical operator information lying outside a finite admissible logical representability subsystem. Thus the confinement mechanism is realized entirely within the logical structure of encoded quantum information.

Throughout this example we consider a general quantum code specified by an encoding isometry. No particular coding architecture is assumed. In particular, the construction is independent of stabilizer codes, topological codes, tensor-network codes, or any other specific realization. Such realizations will be discussed only as illustrative special cases after the abstract obstruction has been established.

Physical realization. Let

$$\mathcal{H}_N = \bigotimes_{j=1}^{N} \mathcal{H}_j$$

be the physical Hilbert space, where every local Hilbert space $\mathcal{H}_j$ is finite dimensional. The logical information is encoded in a finite-dimensional logical Hilbert space

$$\mathcal{H}_L,$$

through an isometric encoding map

$$V : \mathcal{H}_L \longrightarrow \mathcal{H}_N,$$

satisfying

$$V^\dagger V = \mathbf{1}_{\mathcal{H}_L}.$$

The corresponding encoded code subspace is

$$\mathcal{C} = V(\mathcal{H}_L) \subseteq \mathcal{H}_N.$$

No assumption is made concerning the internal structure of the encoding. The encoding isometry is treated as part of the physical realization, while the admissible dynamics acting on the encoded states satisfy Axioms A1–A6.

The admissible dynamics satisfy Axioms A1–A6.

Unlike Example 1, no assumption is made concerning GHZ-type coherence or memory-mediated generation of multipartite correlations. Unlike Example 2, no tensor-network realization is assumed. The only structural restriction considered here is finite logical representability.

We assume that the admissible initial states belong to the encoded logical state class

$$\mathfrak{S}_{\text{code}} \subseteq \mathfrak{S}(\mathcal{H}_N),$$

defined by

$$\mathfrak{S}_{\text{code}} = \left\{ \rho = V \rho_L V^\dagger \ : \ \rho_L \in \mathfrak{S}(\mathcal{H}_L) \right\}.$$

Thus every admissible physical state is the image of a logical state under the encoding isometry.

The present realization concerns the structural properties of these encoded states rather than the physical process by which the encoding is produced. Accordingly, no assumption is made that the admissible dynamics generate the code space from an unencoded initial state. Instead, the code manifold serves as the natural state class on which the abstract confinement mechanism is realized.

Logical entanglement-structure space. The relevant structural objects in the present realization are not defined by the choice of encoding map alone, but by the logical information structures preserved by the encoded states. Accordingly, we introduce the logical entanglement-structure space

$$\mathfrak{E}_{\text{code}},$$

whose elements classify encoded states according to the admissible logical structural information relevant for the representability scale under consideration.

The corresponding entanglement-structure map is written as

$$\mathcal{E} : \mathfrak{S}(\mathcal{H}_N) \longrightarrow \mathfrak{E}_{\text{code}}.$$

The structural equivalence relation is defined by

$$\rho_1 \sim_{\text{code}} \rho_2$$

whenever the two encoded states determine the same logical operator sector and the same logical protection data retained by the chosen structural description.

Equivalently,

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2)$$

if and only if the two states are indistinguishable with respect to the primitive logical structural information defining the present realization. The quotient

$$\mathfrak{E}_{\text{code}} = \mathfrak{S}(\mathcal{H}_N) / \sim_{\text{code}}$$

therefore classifies encoded states according to their logical structural content rather than according to the values of an a priori collection of structural observables.

Importantly, the structural space $\mathfrak{E}_{\text{code}}$ is not identified with a single coding parameter such as code distance, logical qubit number, or stabilizer weight distribution. Instead, such quantities arise as particular coordinates or specializations of the more general logical structure.

We introduce the logical protection functional

$$\mathcal{P}_L : \mathfrak{E}_{\text{code}} \longrightarrow \mathbb{R}_{\geq 0},$$

which quantifies the amount of logical operator structure that must be retained in order to faithfully characterize the encoded information.

For an encoded state

$$\eta = \mathcal{E}(\rho),$$

the quantity

$$\mathcal{P}_L(\eta)$$

measures the structural scale of logical protection associated with the entanglement structure η.

The precise realization of $\mathcal{P}_L$ depends on the chosen code family and logical operator structure. In the general setting considered here, it is treated as a structural coordinate on $\mathfrak{E}_{\text{code}}$ rather than as an independent dynamical quantity.

For stabilizer-code realizations, the functional $\mathcal{P}_L$ admits the familiar specialization in terms of logical operator support. In that case, the code distance provides a concrete realization of the logical protection scale. However, the confinement mechanism developed below depends only on the existence of the abstract structural coordinate $\mathcal{P}_L$, not on any particular code construction.

The purpose of the present example is therefore not to establish a restriction on a specific family of quantum codes, but to show that finite logical representability imposes a structural constraint on protected logical information for code families satisfying the logical representability growth condition.

Logical representability subsystem. The representability structure is now specified through a finite-scale logical operator subsystem. The purpose is not to identify the representable sector with all operators of bounded physical weight. Rather, the finite scale D defines a collection of logically accessible observables whose generated operator subsystem satisfies the abstract requirements of Axiom A5.

Let

$$V_D \subseteq \mathcal{B}(\mathcal{H}_N)$$

denote the linear span of all admissible logical observables accessible at representability scale D. These observables are generated by logical operators whose physical implementation remains within the prescribed accessibility constraint.

The corresponding representable operator subsystem is defined as

$$\mathcal{A}_D^{(\text{code})} := \operatorname{Alg}(V_D),$$

where $\operatorname{Alg}(V_D)$ denotes the unital operator subsystem generated by the accessible logical observables.

Importantly,

$$\mathcal{A}_D^{(\text{code})} \neq \{X \in \mathcal{B}(\mathcal{H}_N) : \operatorname{wt}(X) \leq D\}.$$

The representable subsystem is therefore not defined by a naive truncation of physical operator weight. Instead, it contains precisely the logical operator information retained at scale D, together with the algebraic closure required by the abstract representability framework.

For the present realization, we assume that

$$\mathcal{A}_D^{(\mathrm{code})}$$

satisfies the operator-system requirements of Axiom A5. We further assume that the accessible logical operator subsystem

$$\mathcal{A}_D^{(\mathrm{code})}$$

is compatible with the locality structure imposed by Axiom A2, so that the logical observables retained at representability scale D respect the admissible notion of locality used throughout the framework.

In particular, it is assumed to be a unital operator subsystem of

$$\mathcal{B}(\mathcal{H}_N)$$

belonging to the class of admissible representability subsystems covered by Axiom A6. Consequently, the existence of a completely positive, unital, idempotent representability projection

$$\Pi_D^{(\mathrm{code})} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{(\mathrm{code})}$$

is not derived from the construction of the accessible logical operator subsystem itself. Rather, it is an additional realization assumption inherited from Axiom A6, and the present logical-code realization is restricted to representability subsystems for which such a projection exists.

Accordingly, the representability map is defined by the completely positive unital idempotent projection

$$\Pi_D^{(\mathrm{code})} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{(\mathrm{code})},$$

satisfying

$$\left(\Pi_D^{(\mathrm{code})}\right)^2 = \Pi_D^{(\mathrm{code})}.$$

The map

$$\Pi_D^{(\mathrm{code})}$$

therefore extracts the component of an arbitrary physical operator that remains accessible within the finite logical representability scale.

The existence of such a completely positive unital idempotent projection is part of the admissible representability structure postulated by Axiom A6. The present realization therefore does not claim that every logical operator subsystem generated from accessible observables necessarily admits such a projection. Rather, it considers only those logical representability subsystems satisfying the operator-system compatibility assumptions required by Axiom A6.

This construction should be interpreted as a logical representability restriction rather than as a restriction on the existence of logical operators themselves. Logical operators of arbitrarily large physical support may exist in the full Hilbert space, but they are not necessarily contained within the finite-scale representable subsystem $\mathcal{A}_D^{(\mathrm{code})}$.

The logical representability defect associated with the admissible logical operator subsystem is defined by restricting attention to logical observables whose information content is subject to the finite representability scale. Let

$$\mathcal{A}_{\log} = V\mathcal{B}(\mathcal{H}_L)V^\dagger$$

denote the encoded logical operator algebra. The logical representability defect is defined by

$$O_L(\rho) := \sup_{\substack{X\in\mathcal{A}_{\log} \\ \|X\|\le 1}} \left| \mathrm{Tr}\left[X\rho\right] - \mathrm{Tr}\left[\Pi_D^{(\mathrm{code})}(X)\rho\right] \right|.$$

Equivalently,

$$O_L(\rho) = \sup_{\substack{X \in \mathcal{A}_{\log} \\ \|X\| \le 1}} \left| \mathrm{Tr}\left[\left(X - \Pi_D^{(\mathrm{code})}(X) \right) \rho \right] \right|.$$

Thus O_L measures the portion of logical operator information that is not retained within the finite logical representability subsystem $\mathcal{A}_D^{(\mathrm{code})}$, rather than the loss of arbitrary physical operator information. Equivalently,

$$O_L(\rho) = \sup_{\|X\| \le 1} \left| \mathrm{Tr}\left[(X - \Pi_D^{(\mathrm{code})}(X))\rho \right] \right|.$$

This definition measures the maximal discrepancy between the expectation value of an accessible logical observable and the expectation value retained after projection onto the finite logical representability subsystem. Therefore, the defect quantifies inaccessible logical operator information rather than a general distance between physical states and their projected descriptions.

Unlike a trace-distance estimate between Schrödinger-picture states, this definition is directly formulated in terms of the operator subsystem appearing in Axiom A6. Therefore it is compatible with the representability structure specified by the confinement framework.

As in the previous realizations, the corresponding structural functional is

$$\mathcal{F}_{O_L} : \mathfrak{E}_{\mathrm{code}} \longrightarrow \mathbb{R},$$

defined through

$$\mathcal{F}_{O_L}(\mathcal{E}(\rho)) = O_L(\rho).$$

The relationship between the logical representability defect and the structural quotient is established in Proposition 7, where it is shown that the induced functional is constant on the fibres of $\mathcal{E}$ under the assumptions defining the present realization.

Logical distance–representability compatibility. The confinement mechanism is obtained only for logical realizations in which the logical protection coordinate controls the amount of information lost under finite representability. Accordingly, the following condition is not a consequence of quantum error correction alone, nor of Axioms A1–A6. It is an additional realization hypothesis specifying when logical distance provides a valid structural coordinate for the representability obstruction.

Accordingly, we impose the following code-dependent compatibility condition.

Assumption: Logical representability growth.

Assumption 1. *For every representability scale D, there exists a nondecreasing function*

$$g_D : \mathbb{N} \to \mathbb{R}_{\ge 0}$$

with

$$g_D(d) \to \infty \qquad \text{as} \qquad d \to \infty,$$

such that every logical entanglement structure

$$\eta \in \mathfrak{E}_{\mathrm{code}}$$

satisfies

$$\mathcal{F}_{O_L}(\eta) \ge g_D(d_L(\eta)).$$

Equivalently, logical structures requiring larger logical operator support necessarily contain a larger amount of information outside the finite representability subsystem $\mathcal{A}_D^{(\mathrm{code})}$.

This assumption is a realization-dependent structural criterion specifying when logical protection and finite representability become quantitatively incompatible. It is not claimed to follow from quantum error correction alone, nor from Axioms A1–A6. Rather, it identifies the class of logical-code realizations for which increasing logical protection necessarily requires increasing amounts of logical operator information outside the finite-scale representable subsystem $\mathcal{A}_D^{(\mathrm{code})}$.

Accordingly, the confinement result should not be interpreted as a universal restriction on quantum error-correcting codes. Instead, it establishes that any code family satisfying the logical representability growth condition exhibits a finite-representability obstruction on sufficiently protected logical structures. The role of the present assumption is therefore to characterize the compatibility relation between logical protection and representability, rather than to encode the confinement conclusion itself.

Assumption: Logical realization of Axiom A6. This assumption specifies the realization map between the abstract Axiom A6 compatibility estimate and the observable O_L.

Assumption 2. *For the logical representability subsystem*

$$\Pi_D^{(\mathrm{code})} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{(\mathrm{code})},$$

the representability compatibility estimate of Axiom A6 is assumed to induce the corresponding observable-level bound

$$O_L(\rho(T)) \le \varepsilon_D(T)$$

for every admissible evolution. Equivalently, the finite representability error controlling the admissible logical subsystem also bounds the maximal loss of expectation values measured by the logical representability defect.

This assumption specifies how the abstract compatibility estimate of Axiom A6 is realized for the present logical-code construction. It is therefore realization-dependent and is not claimed to follow from Axioms A1–A6 alone.

Assumption: Fibre invariance of the logical representability defect.

Assumption 3. *The structural equivalence relation defining $\mathfrak{E}_{\mathrm{code}}$ is assumed to preserve the logical representability defect. Specifically, whenever two encoded states belong to the same structural equivalence class,*

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2),$$

their logical representability defects agree,

$$O_L(\rho_1) = O_L(\rho_2).$$

Equivalently, the structural quotient identifies precisely those encoded states that are indistinguishable with respect to the expectation values entering the definition of O_L. This condition ensures that the functional O_L, initially defined on the physical state space $\mathfrak{S}(\mathcal{H}_N)$, descends to a well-defined functional on the logical entanglement-structure space $\mathfrak{E}_{\mathrm{code}}$.

Logical representability structural observable. The logical representability defect defined above provides the structural observable associated with the finite-scale logical operator subsystem. The observable

$$O_L$$

quantifies the logical operator information that cannot be reconstructed from the accessible representable subsystem

$$\mathcal{A}_D^{(\mathrm{code})}$$

when restricted to the encoded logical operator algebra

$$\mathcal{A}_{\log}.$$

It therefore measures a logical representability obstruction rather than a general physical-state representability defect. It is not identified with a code distance or with a logical operator weight. Rather, it measures the loss of logical structural information induced by the finite representability projection.

The induced functional $\mathcal{F}_{O_L}$ therefore defines a structural observable on the code-adapted entanglement-structure space $\mathfrak{E}_{\text{code}}$.

By Assumption 3, the logical representability defect is constant on the fibres of the structural map. Therefore,

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2)$$

implies

$$O_L(\rho_1) = O_L(\rho_2).$$

Hence the induced functional $\mathcal{F}_{O_L}$ is well defined on the logical entanglement-structure space.

Proposition: Realization of an admissible logical structural observable.

Proposition 7. *For the quantum-code realization defined above, suppose that the following construction conditions hold:*

(i) the logical structural equivalence relation preserves the accessible logical operator data required to evaluate the representability defect O_L;

(ii) the projection $\Pi_D^{(\text{code})}$ satisfies the assumptions of Axiom A5;

(iii) the observable-level compatibility estimate of Assumption 2 holds.

Then the functional O_L satisfies all requirements of Definition 10.. Then the logical representability-defect functional

$$O_L(\rho) = \sup_{\|X\|\leq 1} \left| \mathrm{Tr}(X\rho) - \mathrm{Tr}\left(\Pi_D^{(\text{code})}(X)\rho\right) \right|$$

defines an admissible structural observable,

$$O_L \in \mathcal{O}_{\text{str}}.$$

Proof. We verify explicitly that the logical-code realization satisfies each condition appearing in Definition 10. The verification uses both the construction of the structural quotient and the additional realization condition specified by Assumption 2.

First, fibre constancy follows from the defining properties of the logical structural quotient together with the construction of the representability projection. If

$$\mathcal{E}(\rho_1) = \mathcal{E}(\rho_2),$$

then the two encoded states possess identical logical structural data, including the logical operator information retained within the admissible representability subsystem $\mathcal{A}_D^{(\text{code})}$. Since the quantity

$$O_L(\rho) = \sup_{\|X\|\leq 1} \left| \mathrm{Tr}(X\rho) - \mathrm{Tr}\left(\Pi_D^{(\text{code})}(X)\rho\right) \right|$$

depends only on this retained logical operator information, it follows that

$$O_L(\rho_1) = O_L(\rho_2).$$

Hence O_L is constant on the fibres of the structural map.

Second, locality compatibility is part of the realization of the logical representability subsystem rather than a consequence of the existence of the projection

$$\Pi_D^{(\mathrm{code})}.$$

Specifically, we assume that the accessible logical operator subsystem

$$\mathcal{A}_D^{(\mathrm{code})}$$

is chosen so as to be compatible with the locality structure required by Axiom A2. Under this realization assumption, the logical representability defect is evaluated only on locality-compatible accessible logical operator information, and the locality requirement of Definition 10 is therefore satisfied.

Third, memory stability is not a consequence of the absence of an explicit memory variable in the definition of O_L. Rather, it is inherited from the admissible reduced physical evolution. Indeed, the logical representability defect is evaluated on the physical state $\rho(T)$, and any dependence on auxiliary memory degrees of freedom enters only through this reduced state. By Axiom A4, admissible evolutions satisfy the required memory stability condition for the reduced physical dynamics. Consequently, whenever the reduced evolution obeys the stability estimate of Axiom A4, the induced structural observable satisfies the corresponding memory stability requirement.

Fourth, representability compatibility is not an independent consequence of Axioms A1–A6 alone. In the present logical-code realization it is imposed through Assumption 2, which specifies the observable-level form of the Axiom A6 compatibility requirement. Hence,

$$O_L(\rho(T)) \le \varepsilon_D(T)$$

for every admissible evolution, and the logical representability-defect functional satisfies the required finite-scale compatibility estimate.

It remains to verify uniform boundedness. For any logical operator satisfying

$$X \in \mathcal{A}_{\log}, \qquad \|X\| \le 1,$$

we have

$$\begin{aligned} \left|\mathrm{Tr}(X\rho) - \mathrm{Tr}\left(\Pi_D^{(\mathrm{code})}(X)\rho\right)\right| &= \left|\mathrm{Tr}\left[\left(X - \Pi_D^{(\mathrm{code})}(X)\right)\rho\right]\right| \\ &\le \left\|X - \Pi_D^{(\mathrm{code})}(X)\right\|. \end{aligned}$$

Since $\Pi_D^{(\mathrm{code})}$ is completely positive and unital, it is contractive in the operator norm. Therefore,

$$\left\|\Pi_D^{(\mathrm{code})}(X)\right\| \le \|X\| \le 1,$$

and hence,

$$\left\|X - \Pi_D^{(\mathrm{code})}(X)\right\| \le 2.$$

Taking the supremum over the logical operator sector gives

$$0 \le O_L(\rho) \le 2.$$

Therefore the logical representability-defect functional is uniformly bounded.

Therefore every requirement in Definition 10 is satisfied within the present logical-code realization. Consequently,

$$O_L \in \mathcal{O}_{\mathrm{str}}.$$

□

Logical protection scale as a structural coordinate. The preceding construction identifies the finite-scale logical representability defect as an admissible structural observable. We now associate the obstruction with a logical protection scale. Importantly, this quantity is not identified with the distance of a fixed code, since code distance is a property of the encoding structure rather than of an individual encoded state.

For a logical entanglement structure

$$\eta \in \mathfrak{E}_{\mathrm{code}},$$

let

$$\mathcal{L}_\eta$$

denote the logical operator sector required to characterize the structural information associated with η. We define the logical protection coordinate

$$d_L : \mathfrak{E}_{\mathrm{code}} \longrightarrow \mathbb{N}$$

by

$$d_L(\eta) = \min \{\mathrm{wt}(L) : L \in \mathcal{L}_\eta, \quad L \notin \mathbb{C}\mathbf{1}\}.$$

Thus $d_L(\eta)$ measures the minimum physical support required by the logical operator information associated with the structural class η. It is therefore a structural coordinate describing the operator complexity of a logical sector, rather than the intrinsic distance parameter of a fixed quantum code.

For a given code family, the standard code distance provides a global lower bound on the protection scale of all nontrivial logical operators. However, the confinement argument below concerns the structural coordinate $d_L(\eta)$, whose variation across logical structures determines the amount of representability information required.

Importantly, the logical distance is not introduced as an additional dynamical assumption. It is a structural coordinate on the entanglement structure space, analogous to other structural quantities appearing in Sections 1–4. The confinement mechanism arises from the representability restriction imposed by Axioms A5–A6.

Representable logical scale. The finite-scale logical subsystem

$$\mathcal{A}_D^{(\mathrm{code})}$$

contains only the logical operator information accessible at representability scale D. The corresponding admissible evolution preserves this information only up to the representability error prescribed by Axiom A6.

Using Assumption 1, the maximal representable logical distance is defined by

$$d_{\max}(T, D) := \max \{d \in \mathbb{N} : g_D(d) \le \varepsilon_D(T)\}.$$

Because g_D is nondecreasing, the admissible set

$$\{d \in \mathbb{N} : g_D(d) \le \varepsilon_D(T)\}$$

is an initial interval of $\mathbb{N}$. Therefore the maximum exists whenever the representable logical scales are bounded, and it satisfies

$$g_D(d) \le \varepsilon_D(T) \quad \Longrightarrow \quad d \le d_{\max}(T, D).$$

Equivalently, $d_{\max}(T, D)$ is the largest logical protection scale for which the required representability information remains below the admissible compatibility error of the finite logical subsystem.

Equivalently, every logical entanglement structure satisfying

$$d_L(\eta) > d_{\max}(T, D)$$

necessarily obeys

$$O_L(\eta) > \varepsilon_D(T).$$

The quantity $d_{\max}(T, D)$ therefore depends on the chosen representability scale together with the quantitative compatibility bound of Axiom A6, rather than on any particular realization of a quantum code.

Logical representability obstruction. By Assumption 2,

$$O_L(\rho(T)) \leq \varepsilon_D(T)$$

for every admissible evolution.

Thus every reachable logical entanglement structure satisfies the representability constraint encoded by the finite logical subsystem.

Proposition: Logical representability obstruction.

Proposition 8. *For every admissible evolution satisfying Axioms A1–A6,*

$$d_L\left(\mathcal{E}(\rho(T))\right) \leq d_{\max}(T, D).$$

Consequently, every logical entanglement structure

$$\eta \in \mathfrak{E}_{\text{code}}$$

satisfying

$$d_L(\eta) > d_{\max}(T, D)$$

cannot belong to the reachable structural set

$$\mathfrak{R}_{\mathcal{E}}(T).$$

Proof. Let

$$\eta = \mathcal{E}(\rho(T))$$

be a reachable logical structure generated by an admissible evolution. By Assumption 2,

$$\mathcal{F}_{O_L}(\eta) = O_L(\rho(T)) \leq \varepsilon_D(T).$$

The logical distance–representability compatibility condition of Assumption 1 gives

$$\mathcal{F}_{O_L}(\eta) \geq g_D(d_L(\eta)).$$

Therefore,

$$g_D(d_L(\eta)) \leq \varepsilon_D(T).$$

By the definition of $d_{\max}(T, D)$,

$$d_L(\eta) \leq d_{\max}(T, D).$$

Hence every reachable logical structure satisfies

$$d_L(\eta) \leq d_{\max}(T, D).$$

Conversely, if

$$d_L(\eta) > d_{\max}(T, D),$$

then monotonicity of g_D implies

$$g_D(d_L(\eta)) > \varepsilon_D(T),$$

and therefore

$$\mathcal{F}_{O_L}(\eta) > \varepsilon_D(T).$$

Such a structure cannot satisfy the Axiom A6 representability compatibility condition and therefore cannot belong to the reachable structural region.

□

Stabilizer-code specialization. The preceding construction applies to general encoded logical structures. A standard explicit realization is provided by stabilizer quantum error correcting codes. The present specialization does not modify the abstract logical representability framework; rather, it provides a concrete setting in which the logical operator structure and protection scale admit a direct interpretation.

Let

$$V : \mathcal{H}_L \longrightarrow \mathcal{H}_N$$

denote an encoding isometry from a logical Hilbert space $\mathcal{H}_L$ into the physical Hilbert space of N qubits. The encoded code space is

$$\mathcal{C} = V(\mathcal{H}_L).$$

For a stabilizer code with stabilizer group

$$\mathcal{S},$$

this subspace may equivalently be characterized as

$$\mathcal{C} = \{|\psi\rangle : S|\psi\rangle = |\psi\rangle, \quad \forall S \in \mathcal{S}\}.$$

The corresponding admissible physical state class is therefore

$$\mathfrak{S}_{\text{code}} = \left\{\rho = V\rho_L V^\dagger : \rho_L \in \mathfrak{S}(\mathcal{H}_L)\right\}.$$

The logical operator algebra embedded into the physical Hilbert space is

$$\mathcal{A}_{\log} = V\mathcal{B}(\mathcal{H}_L)V^\dagger.$$

Within this specialization, the logical operator sector is determined by the normalizer

$$N(\mathcal{S}),$$

and the standard stabilizer-code distance is defined by

$$d_{\text{stab}} = \min\{\text{wt}(P) : P \in N(\mathcal{S}) \setminus \mathcal{S}\}.$$

For stabilizer-code realizations, the abstract structural coordinate $d_L(\eta)$ is not identified with the stabilizer code distance. Rather, the stabilizer distance provides a code-level protection parameter that characterizes the minimum support of any nontrivial logical operator in the code.

The stabilizer distance is fixed once the stabilizer code is specified. By contrast, $d_L(\eta)$ describes the protection scale associated with the particular logical operator sector required by the structural class η.

For stabilizer-code realizations, the equivalence class of logical operators associated with a structural class η is represented by elements of the normalizer quotient

$$N(\mathcal{S})/\mathcal{S}.$$

The corresponding operator representatives form the logical sector $\mathcal{L}_\eta$ used in the definition of $d_L(\eta)$. Therefore, the minimum support required by a particular structural logical sector satisfies

$$d_L(\eta) \geq d_{\mathrm{stab}},$$

because d_{stab} is the minimum support over all nontrivial logical operators in the code. The quantity d_{stab} therefore provides the code-wide lower protection scale, whereas $d_L(\eta)$ records the protection requirement of a specific logical structural sector.

The confinement argument therefore remains a statement about the structural coordinate $d_L(\eta)$, with d_{stab} entering only as a code-dependent reference scale and lower-bound parameter.

Equivalently, stabilizer-code logical structures whose required logical protection scale satisfies

$$d_L(\eta) > d_{\max}(T, D)$$

cannot belong to the reachable structural region generated by admissible dynamics at representability scale D.

The stabilizer distance d_{stab} only determines the minimum protection scale of the code family and does not itself determine whether an individual logical structural sector is reachable.

The present confinement mechanism does not depend on stabilizer structure itself. Stabilizer codes provide one explicit realization in which logical operators, code distance, and correctability properties admit a concrete interpretation. The confinement result applies to any code realization that satisfies the logical representability growth condition and the corresponding observable-level realization of Axiom A6.

Representative examples include surface-code constructions, toric-code realizations, quantum low-density parity-check codes, and holographic quantum codes. These families differ in their geometric realization of logical operators, but they share the abstract feature relevant to the present framework: logical information requires operator support whose representability must remain compatible with the finite-scale subsystem

$$\mathcal{A}_D^{(\mathrm{code})}.$$

Interpretation of the logical representability obstruction. The preceding proposition establishes a confinement result for protected logical structures under finite representability constraints. The result should not be interpreted as a no-go theorem for quantum error correction or as a statement that large-distance quantum codes cannot exist.

Rather, the obstruction concerns dynamical reachability within the abstract framework of Sections 1–4. A logical structure may exist as an element of the full kinematic entanglement-structure space while remaining outside the reachable structural region generated by admissible dynamics at finite representability scale.

Equivalently,

$$\text{existence of a logical code structure} \neq \text{dynamical accessibility of that structure.}$$

The confinement mechanism arises because, for code families satisfying Assumptions 1 and 2, the logical information required to support sufficiently protected structures grows beyond the retention capacity of the finite representable operator subsystem

$$\mathcal{A}_D^{(\mathrm{code})}.$$

Thus the obstruction is a statement about structural realizability rather than about the mathematical existence of quantum error-correcting codes.

Independence from the GHZ realization. The logical representability obstruction is structurally distinct from the GHZ realization of Example 1.

In the GHZ construction, the obstruction was associated with the inability of finite memory dynamics to preserve the global coherence witness

$$X_{\mathrm{GHZ}}.$$

The relevant limitation arose from the combined effects of memory mixing and representability compatibility.

By contrast, the present realization does not require:

- a GHZ coherence witness,
- a memory-mediated generation mechanism,
- or any special multipartite coherence structure.

The obstruction here is generated entirely by the finite logical representability subsystem. The relevant structural quantity is the logical distance

$$d_L(\eta),$$

which measures the protection scale of the logical information rather than the depth of multipartite coherence.

Therefore the two realizations represent genuinely different mechanisms:

$$\text{GHZ coherence confinement} \neq \text{logical protection confinement.}$$

Independence from the tensor-network realization. The present construction is also distinct from the tensor-network representability realization of Example 2.

In the tensor-network example, the representability restriction was imposed through an MPO-adapted operator subsystem

$$\mathcal{A}_D^{\mathrm{MPO}},$$

and the obstruction concerned structural information that could not be faithfully retained within a finite tensor-network description.

Here, the representability subsystem

$$\mathcal{A}_D^{(\mathrm{code})}$$

is instead generated by accessible logical observables. The excluded structures are not defined by tensor-network complexity or approximation quality, but by the inability to represent sufficiently protected logical operator sectors.

Consequently,

$$\text{tensor-network representability} \neq \text{logical representability.}$$

Both realizations instantiate the same abstract confinement theorem, but they identify different structural coordinates and different forbidden families.

Compatibility with locality. The logical representability obstruction is compatible with locality-based quantum dynamics. In particular, the result does not assert that logical structures are forbidden because information must propagate outside a causal region.

The admissible dynamics continue to satisfy the locality requirements of Axiom A2. The obstruction appears only after imposing the additional representability constraint of Axioms A5–A6.

Thus even when locality permits the formation of a logical structure, finite representability may prevent the corresponding protected information from remaining encoded within the admissible logical subsystem.

The confinement mechanism is therefore distinct from a purely locality-based obstruction. It arises from the incompatibility between a target structural requirement and the finite operator information retained by the representability map

$$\Pi_D^{(\mathrm{code})}.$$

Remark: quantum-code realizations. The general construction applies to any encoded logical structure satisfying the representability assumptions of Axioms A5–A6. Stabilizer codes provide a standard explicit realization, where

$$d_L = \min \{\mathrm{wt}(P) : P \in N(S) \setminus S\}.$$

The same structural mechanism can therefore be considered in a variety of code families, including surface-code constructions, toric-code realizations, quantum low-density parity-check codes, and holographic code models.

The role of these examples is not to derive the confinement theorem from a particular code family. Rather, they illustrate different physical realizations of the same abstract principle:

finite representability constrains the dynamically reachable region of structural space.

Conclusion of the logical-code realization. The quantum-code realization therefore provides a third independent instantiation of the abstract structural confinement theorem.

The representability projection

$$\Pi_D^{(\mathrm{code})} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{(\mathrm{code})}$$

induces the structural inequality

$$d_L(\eta) \leq d_{\max}(T, D)$$

for all reachable structures

$$\eta \in \mathfrak{R}_{\mathcal{E}}(T).$$

Equivalently,

$$\boxed{d_L(\eta) > d_{\max}(T, D) \quad \Longrightarrow \quad \eta \notin \mathfrak{R}_{\mathcal{E}}(T)}$$

and the corresponding family of highly protected logical structures lies outside the dynamically reachable structural region.

This completes the logical-code realization of the confinement hull of Theorem 3.

Accessible logical operator realization. Within the stabilizer realization, the representability scale D is interpreted as an accessibility scale for logical information. In particular, let

$$\mathcal{A}_D^{(\mathrm{stab})} \subseteq \mathcal{A}_{\log}$$

denote the admissible logical operator subsystem generated by logical operators admitting representatives within the prescribed accessibility scale D.

The construction does not identify $\mathcal{A}_D^{(\mathrm{stab})}$ with the set of all logical operators of weight bounded by D. Such a set need not itself possess the operator subsystem structure required by Axiom A5. Instead, the present realization selects the operator subsystem generated by accessible logical operators and assumes that it satisfies the completely positive unital idempotent representability requirements of Axioms A5–A6.

The associated representability projection is therefore

$$\Pi_D^{(\mathrm{stab})} : \mathcal{B}(\mathcal{H}_N) \longrightarrow \mathcal{A}_D^{(\mathrm{stab})},$$

with

$$\left(\Pi_D^{(\mathrm{stab})}\right)^2 = \Pi_D^{(\mathrm{stab})}.$$

The obstruction derived in the preceding construction consequently applies whenever logical structures require operator information outside this admissible logical subsystem.

Representative code families. The preceding construction is independent of any particular code family. Nevertheless, several important families of quantum error-correcting codes provide explicit realizations of the logical representability structure.

Surface and toric codes. For surface-code and toric-code constructions, logical operators are represented by extended string or loop operators. The logical distance corresponds to the minimal support required for a nontrivial logical operator.

Within the present framework, increasing logical distance corresponds to an increasing requirement for representable logical operator information. A finite representability scale therefore restricts the class of logical structures whose protection properties can be faithfully retained.

Quantum LDPC codes. Quantum low-density parity-check (LDPC) codes provide families in which large code distance is compatible with sparse stabilizer structure. The present obstruction applies to such codes because it is formulated in terms of logical representability rather than a particular geometric realization.

Thus, even when locality and sparse interaction constraints are satisfied, logical structures requiring inaccessible operator information remain outside the reachable structural region.

Holographic codes. Holographic code constructions provide another realization in which logical information is encoded through highly nontrivial mappings between logical and physical degrees of freedom.

The confinement mechanism remains unchanged: the obstruction does not arise from the geometric form of the encoding, but from the finite logical representability subsystem selected by the admissible scale D.

Compatibility with locality. The logical representability obstruction is compatible with the locality requirements imposed by Axiom A2. In particular, the exclusion of highly protected logical structures does not rely on any violation of causal propagation bounds or on faster-than-allowed information transfer.

Rather, the obstruction arises because the logical information defining such structures cannot be retained within the admissible finite logical representability subsystem. The limitation is therefore structural rather than causal.

Consequently, a target logical entanglement structure may remain compatible with locality and finite-time dynamics while nevertheless lying outside the reachable structural region

$$\mathfrak{R}_{\mathcal{E}}(T)$$

because its defining logical operator information exceeds the representability capacity permitted by

$$\mathcal{A}_D$$

and the associated representability compatibility condition

$$g_D(d_L(\eta)) \leq \varepsilon_D(T).$$

Equivalently,

$$g_D(d_L(\eta)) > \varepsilon_D(T)$$

identifies a family of logical structures whose required logical operator information exceeds the admissible finite representability threshold. Consequently, such structures cannot be dynamically realized within the admissible finite representability scale.

This completes the third realization of the abstract structural confinement theorem under the logical representability growth condition. Unlike the GHZ realization, which is controlled by multipartite coherence preservation, and the tensor-network realization, which is controlled by finite operator representation, the present realization identifies finite logical representability as the mechanism generating confinement within the class of logical structures satisfying the stated compatibility assumptions.

6 Discussion

The results developed throughout this work establish structural confinement as a general mechanism for deriving dynamical accessibility constraints in quantum structure space. The central result is not the exclusion of any particular family of states, but the identification of a general relation between representability and structural reachability.

The confinement theorem shows that once a structural space, representability subsystem, and admissible structural observable have been specified, the corresponding reachable region is constrained by the compatibility properties of the representability map. Structures whose defining information exceeds the admissible representability scale are excluded from the reachable structural region, even when they remain valid elements of the full kinematic state space.

6.1 Structural confinement as an accessibility principle

The examples developed above suggest that structural confinement should be viewed as an accessibility principle rather than as a family of isolated no-go statements.

Traditional impossibility results in quantum theory often identify constraints on evolution, preparation, measurement, or computational resources. The present framework introduces a complementary viewpoint: constraints arise because the information required to specify a target structure may lie outside the admissible representational sector preserved by the dynamics.

This distinction can be summarized as

$$\text{kinematic existence} \neq \text{dynamical reachability} \neq \text{representational accessibility}.$$

The first relation concerns whether a structure exists in the full Hilbert space. The second concerns whether admissible dynamics can generate that structure. The third concerns whether the structural information defining that structure can be retained within the allowed representability framework.

Structural confinement arises precisely from the failure of the third condition. A target structure may therefore be compatible with locality, finite-time evolution, and the underlying Hilbert-space description, while still remaining outside the dynamically reachable structural region.

This perspective provides a unifying interpretation of the preceding realizations: GHZ coherence, tensor-network structure, and protected logical information are different examples of the same general phenomenon in which finite representability induces a boundary on structural accessibility.

6.2 Conceptual distinction from existing quantum constraints

The structural confinement framework differs from existing constraint principles in quantum information because the source of the obstruction is not identified with a specific dynamical limitation.

Locality-based results constrain the propagation of information through spacetime or interaction graphs. Complexity-based results constrain the resources required to prepare, approximate, or describe quantum states. Resource-theoretic approaches quantify the availability of particular physical resources. Error-correction theory characterizes the protection and recovery of encoded information.

The present framework addresses a different question:

Which structural sectors remain dynamically accessible when only a finite representable operator sector is ret

The obstruction therefore does not arise because a structure violates locality, because entanglement generation is impossible, or because a state requires a particular preparation complexity. Rather, the obstruction arises because the information defining the structure cannot be preserved inside the admissible representability subsystem.

This introduces representability as an additional layer of physical constraint. The resulting hierarchy may be expressed as

$$\text{kinematic possibility} \to \text{dynamical accessibility} \to \text{representational accessibility}.$$

Structural confinement identifies the final transition as a genuine source of forbidden regions in quantum structural space.

6.3 Universality across distinct physical realizations

The three realizations developed in Section 5 demonstrate that structural confinement is not tied to a particular physical notion of complexity or structure.

The GHZ realization shows that finite memory and representability constraints can restrict access to global coherence sectors. The tensor-network realization shows that finite operator representability can exclude structures whose information content exceeds an admissible tensor-network scale. The quantum error-correction realization shows that protected logical structures may become inaccessible when their defining logical operator information exceeds the available representability scale.

These examples are not independent applications of unrelated bounds. They are different coordinate realizations of the same abstract mechanism:

$$(\text{structural space}, \text{representability map}, \text{structural observable}) \longrightarrow \text{reachable structural region}.$$

The invariant object is therefore not a particular entanglement measure, tensor-network parameter, or code property. The invariant object is the representability-induced geometry of accessible and inaccessible structural sectors.

6.4 Structural observables and the geometry of reachable regions

A central feature of the framework is that confinement is determined relative to a chosen structural observable rather than by a universal scalar measure of quantum complexity.

Different physical questions naturally select different structural spaces. Consequently, the same dynamical system may exhibit different confinement regions depending on which structural information is regarded as relevant.

The GHZ coherence functional, tensor-network representability defect, and logical representability defect therefore do not compete as alternative measures of complexity. They represent different projections of the same underlying accessibility problem.

This viewpoint suggests that quantum structural analysis should not be limited to determining the amount of a resource contained in a state. A complementary question is whether the information defining that resource remains accessible under a specified representational framework.

6.5 Implications and limitations

The structural confinement theorem provides a general method for deriving forbidden structural regions, but its conclusions depend explicitly on the choice of representability subsystem and structural observable.

The framework therefore does not assert that excluded structures are fundamentally impossible. Rather, it identifies structures that cannot be reached within a specified dynamical and representational regime.

In particular,

$$\text{structural exclusion} \neq \text{kinematic nonexistence}.$$

A structure may exist in the full Hilbert space and may even be compatible with the locality assumptions of the dynamics, while remaining inaccessible because the required structural information lies outside the admissible representable sector.

This distinction is essential for interpreting the confinement mechanism. The framework does not replace existing locality, complexity, or error-correction theories. Instead, it provides an additional structural layer through which their accessibility consequences may be analyzed.

7 Scope, Extensions, and Future Directions

The preceding examples demonstrate that the structural confinement theorem is not a collection of independent impossibility statements associated with different physical settings. Rather, they establish a common accessibility principle: once a physically relevant notion of structure is specified, finite representability constraints induce a nontrivial geometry of dynamically reachable and unreachable sectors within the corresponding structural space.

The central conceptual shift introduced by the framework is that structural existence and structural accessibility are distinct notions. A quantum structure may exist within the full kinematic state space while remaining dynamically inaccessible because the operator information required to realize or preserve that structure cannot be retained within the admissible representability subsystem.

Accordingly, the confinement theorem does not classify which structures are mathematically possible. Instead, it determines which structures remain reachable under a specified combination of dynamics, representability, and structural observables.

Equivalently, a single abstract confinement principle generates different impossibility regions depending on what information is regarded as structural. The physical content of the obstruction is therefore not fixed by the theorem alone, but emerges from the interaction between the chosen structural space and the admissible representability scale.

The GHZ realization identifies confinement arising from the inability of finite memory resources to preserve a structural coherence sector. The tensor-network realization identifies confinement arising from finite operator representability. The quantum error-correcting code realization identifies confinement arising from finite logical representability of protected quantum information.

These examples are therefore not merely demonstrations in separate physical domains. They show that structural confinement defines a common mechanism through which qualitatively different notions of quantum structure acquire corresponding forbidden regions. The invariant object across the examples is not the physical structure itself, but the relation

$$(\text{structural space}, \text{representability map}, \text{structural observable}) \longrightarrow \text{reachable region}.$$

.

These examples therefore establish that structural confinement is a representation-independent phenomenon whose physical interpretation depends on the chosen structural observable.

A defining feature of the present framework is that the obstruction is neither a purely causal limitation nor a conventional complexity bound. Locality-based constraints restrict the speed at which information can propagate, while complexity-based constraints restrict the resources required to prepare or describe states. Structural confinement addresses a different question:

Which structural sectors remain dynamically accessible when only a finite representable operator sector is ret

Thus the framework introduces an intermediate layer between kinematic possibility and dynamical evolution: the accessibility of structures is constrained by the information content preserved by the admissible representation. In this sense, structural confinement provides a general mechanism for deriving impossibility regions that are neither excluded by kinematics nor directly forbidden by locality, but become inaccessible because their defining structural information exceeds the allowed representability scale.

7.1 Scope of the framework

The abstract formulation was intentionally constructed to separate universal mathematical ingredients from model-dependent physical realizations. This separation is essential because the primary object of the theory is not a particular quantum resource or physical architecture, but the accessibility structure induced on a space of physical structures. The framework therefore moves the focus from resource quantification to reachability geometry: rather than asking how much of a given resource is present, it asks which structural classes can be dynamically attained under specified representability constraints. The framework requires:

1. a state class describing admissible initial configurations;
2. an entanglement-structure space specifying the relevant equivalence relation on states;
3. a representable operator subsystem satisfying the requirements of Axioms A5–A6;

4. structural observables capable of detecting inaccessible regions of the structural space.

Once these objects are specified, the confinement theorem provides a representation-independent criterion for excluding structures outside the reachable structural region.

This formulation allows the same mathematical architecture to be applied across different areas of quantum theory without identifying structural confinement with any particular physical mechanism such as locality, tensor-network complexity, or error-correction distance.

7.2 Additional possible realizations

The three explicit examples developed in this work are not exhaustive. Several other important quantum structures provide natural settings in which the same confinement mechanism may be investigated.

Topological phases and long-range entanglement. A natural extension concerns topologically ordered phases, where the relevant structural information is encoded not in local observables but in global topological data.

Possible structural observables include:

$$\gamma(\eta),$$

representing topological entanglement characteristics,

or more generally,

$$\mathcal{F}_{\text{top}} : \mathfrak{C}_{\text{top}} \to \mathbb{R},$$

measuring the accessibility of topological order parameters within a finite representability scale.

A representability obstruction of the form

$$\mathcal{F}_{\text{top}}(\eta) > \epsilon_D(T)$$

would identify topological structures whose defining information cannot be retained within the admissible structural subsystem.

Such a realization would connect the present framework to questions involving topological state preparation, long-range entanglement generation, and quantum phase accessibility.

Continuous-variable quantum systems. Although the examples considered in this work use finite-dimensional Hilbert spaces, the abstract construction does not fundamentally depend on this restriction.

Continuous-variable systems provide another possible realization, where the representable subsystem may be defined through finite-resolution observables, bounded-energy sectors, or experimentally accessible measurement algebras.

The corresponding structural observable could quantify the information lost under finite experimental or operational resolution.

Emergent geometric and holographic structures. The logical-code realization already illustrates that the framework naturally applies to holographic encoding structures. More generally, the theory suggests a possible route for studying emergent geometric descriptions in which representability constraints determine which large-scale structures can arise from microscopic dynamics.

In such settings, structural observables may characterize emergent geometric degrees of freedom rather than conventional entanglement measures.

7.3 Representative realizations and structural observables

The abstract confinement principle developed in Sections 1–4 does not select a unique physical notion of structure. Rather, the confinement region depends on the structural coordinates retained by the chosen realization. The following tables summarize explicit realizations of the framework and possible extensions. Each realization specifies the relevant state class, structural space, representability structure, and the induced structural obstruction.

Table 1: Explicit realizations of the abstract confinement framework developed in this work.

Domain	**State class**	**Structural space**	**Representability subsystem**	**Structural observable**	**Forbidden regime**
GHZ / multipartite coherence	$\mathfrak{S}_{\mathrm{prod}}$ with memory-mediated admissible dynamics	$\mathfrak{C}_{\mathrm{GHZ}}$	$\mathcal{A}_D$ with finite accessible degrees of freedom	$\mathcal{F}_{O_{\mathrm{GHZ}}}$	$\mathcal{F}_{O_{\mathrm{GHZ}}} > C_M e^{-T/\tau_M} + \varepsilon_D(T)$
Tensor-network representation	$\mathfrak{S}_{\mathrm{TN}}$ compatible with finite representability scale	$\mathfrak{C}_{\mathrm{TN}}$	$\mathcal{A}_D^{\mathrm{MPO}}$ MPO-adapted operator subsystem	$\mathcal{F}_{O_{\mathrm{TN}}}$	$\mathcal{F}_{O_{\mathrm{TN}}} > \varepsilon_D(T)$
Quantum error correction / logical structures	Encoded logical manifold $\mathfrak{S}_D^{(\mathrm{code})}$	$\mathfrak{C}_{\mathrm{code}}$	$\mathcal{A}_D^{(\mathrm{code})}$ generated by accessible logical operators	$\mathcal{F}_{O_L}$	$d_L(\eta) > d_{\max}(T, D)$

The preceding examples demonstrate that the same abstract confinement principle can generate physically distinct impossibility regions. The obstruction is determined not by a universal measure of complexity, but by the structural information retained by the chosen representability map. The following table lists further possible realizations of the same framework.

Table 2: Possible extensions of the confinement framework to additional physical domains.

Domain	**Structural space**	**Representability structure**	**Structural observable**	**Expected obstruction**
Topological phases	$\mathfrak{C}_{\mathrm{top}}$	Finite algebra of topological logical operators	$\mathcal{F}_{\mathrm{top}}$	Failure of finite-scale topological representability
Continuous-variable systems	$\mathfrak{C}_{\mathrm{CV}}$	Finite-resolution phase-space operator subsystem	$\mathcal{F}_{\mathrm{res}}$	Structures requiring resolution beyond D
Emergent geometric structures	$\mathfrak{C}_{\mathrm{geom}}$	Finite geometric observable algebra	$\mathcal{F}_{\mathrm{geom}}$	Geometric structures outside finite representability scale
Quantum many-body phases	$\mathfrak{C}_{\mathrm{MB}}$	Finite-dimensional effective operator sector	$\mathcal{F}_{\mathrm{MB}}$	Failure of finite effective description
Holographic and emergent code structures	$\mathfrak{C}_{\mathrm{holo}}$	Restricted logical/geometric operator algebra	$\mathcal{F}_{\mathrm{holo}}$	Structures requiring unbounded logical encoding

Together, these realizations emphasize the central message of the framework: the confinement mechanism is universal, while the physical meaning of the forbidden region depends on what is chosen to constitute structure.

The resulting viewpoint is therefore not a catalogue of separate restrictions on GHZ states, tensor networks, or logical codes. Rather, it provides a common mathematical language for describing how finite representability reshapes the space of dynamically accessible quantum structures.

The novelty of the framework lies in identifying representability itself as a source of structural constraints. Once a representable subsystem is specified, the reachable region of structural space becomes an emergent object determined by the interaction between dynamics, representation, and structural observables. In this sense, structural confinement establishes a general principle for deriving inaccessible sectors of quantum structure without requiring that those sectors be forbidden by locality, kinematics, or the absence of entanglement generation mechanisms.

7.4 Future mathematical directions

Several mathematical questions naturally arise from the present framework.

First, a systematic classification of admissible structural observables $\mathcal{O}_{\mathrm{str}}$ would clarify which physical notions of structure admit rigorous confinement bounds.

Second, sharper versions of the confinement theorem may be obtained by replacing uniform representability estimates with problem-dependent information complexity bounds.

Third, it remains an open direction to understand the relationship between structural confinement and other notions of quantum complexity, including circuit complexity, preparation complexity, and resource-theoretic measures.

Finally, extending the framework beyond finite-dimensional operator algebras may reveal whether analogous confinement principles exist for more general operator-algebraic and field-theoretic settings.

8 Conclusion

The present work introduced a general framework for structural confinement in quantum dynamics. The central result is that dynamical accessibility is not determined solely by the physical evolution law, locality constraints, or entanglement generation mechanisms. Rather, it is constrained by the compatibility between the dynamics and the representational structure used to define quantum information.

The structural confinement theorem provides a general method for deriving such constraints. Given a structural space, an admissible representability subsystem, and a corresponding structural observable, the theorem identifies regions of structural space that cannot be reached by admissible dynamics whenever the required information exceeds the representability scale preserved by the evolution.

This viewpoint changes the interpretation of quantum impossibility results. The obstruction need not arise because a target structure violates locality, because entanglement cannot be generated, or because a particular physical resource is insufficient. Instead, a structure may remain fully compatible with the underlying dynamics while being inaccessible because the information required to characterize it cannot be retained within the admissible representational framework.

The three realizations developed in this work demonstrate the generality of this mechanism. The GHZ realization identifies confinement of multipartite coherence under finite memory and representability constraints. The tensor-network realization identifies confinement arising from finite operator representability. The quantum-code realization identifies confinement of protected logical structures under finite logical representability. These examples are not separate impossibility results, but distinct manifestations of a common structural principle.

The resulting perspective suggests that the relevant question in many quantum dynamical problems is not only whether a structure can exist, but whether it can remain dynamically accessible within a specified representational framework. Structural confinement therefore provides a general language for studying the boundary between physically allowed quantum structures and those that, while mathematically well-defined, cannot be realized within a given finite-scale description.

Future developments of this framework may reveal further connections between representability, quantum complexity, emergent structures, and the operational limits of quantum information processing. The central principle remains:

$$\boxed{\text{dynamical accessibility is constrained by structural representability}}$$

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